

Voltage quality and economic activity

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September 18, 2026

Abstract

While voltage problems are believed to be ubiquitous in low-income countries, the impact of voltage quality on the economy is understudied. Novel minute-by-minute customer-level power measurements linked to panel surveys in Ghana reveal systematic under-voltage, damaging equipment and incentivizing protective investments. Customers would pay more for electricity with better voltage. Quasi-random grid improvements raised voltage by 5.5V and reduced bad voltage hours by 44%, reducing appliance damages, with minimal economic impacts after one year. The limited short-run economic impacts seemingly reflect delayed durable capital adjustment, illustrating the complexity of allocating long-term investments. We provide a framework for evaluating power grid investments.

JEL codes: H54, L94, O1

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1 Introduction

Novel sources of spatially and temporally high-frequency data such as satellite imagery, internet-of-things, and mobile phone records have in recent years greatly enhanced researchers’ abilities to assess the technological determinants of economic performance – including the functioning of the energy grid, the bedrock of every modern economy. In addition to studies on electricity pricing and access, among other topics, novel high-resolution data have enabled extensive analyses of the economic cost of power outages.¹ Widespread power outages have caused deaths and economic crises across the world in for example California, Texas, and Ghana (Abeberese, 2019; Abeberese et al., 2021; Svitek, 2024; Wolfram, 2019). This paper uses customer surveys and high-frequency sensor data collected over several years to analyze an understudied feature of modern energy grids: voltage quality. Voltage problems such as fluctuations and persistent sags can damage appliances and limit appliance functionality, imposing economic costs and affecting electricity use. However, these costs are poorly understood due to a lack of high-resolution voltage data in contexts with poor voltage.

In most countries, electric utilities are charged with providing electricity within $\pm 10\%$ of targeted voltage –for example, 120 volts (V) in the US and 230V in Ghana – so that customers can operate machinery and appliances efficiently and without damage (CENELEC, 2006). Utilities in low-resource contexts often fail to meet these standards. Yet, voltage issues continue to be deprioritized in policy and regulatory debates. The United Nations Sustainable Development Goal 7 – “*affordable, reliable, sustainable and modern energy for all*” – does not mention voltage (UN, 2022). World Bank reports often either do not discuss voltage or depend on household self-reports, which can be inaccurate (WB, 2020, 2021).

¹On outages: Allcott et al. (2016), Fisher-Vanden et al. (2015), Gertler et al. (2017), and Hardy and McCasland (2019). On affordability: Borenstein (2012), Burgess et al. (2021), Cong et al. (2022), Lee et al. (2020), and Levinson and Silva (2022). On responsiveness to prices: Deryugina et al. (2020) and Ito (2014). On formality: Jack and Smith (2020) and McRae (2015). On access: Burlig and Preonas (2023), Dinkelman (2011), Gaggl et al. (2021), Lee et al. (2020), Lewis and Severnini (2020), and Lipscomb et al. (2013). On capacity: Burgess et al. (2021) and Ryan (2021).

Understanding the economic costs of poor voltage quality – in particular relative to improvements in access, reliability, or affordability – is crucial to enable resource-constrained utilities to optimally target grid investments.

We analyze 337 million customer-level electricity reliability and voltage measurements, collected from more than 1,000 utility customers over the span of six years via the installation of socket plug-in GridWatch devices in Accra, Ghana. Power quality measurements are complemented with survey data from over 2,000 micro-enterprises and households. The firms in our survey are predominantly small-scale businesses – such as retail shops, personal care services, food and beverage vendors, and clothing manufacture and repair – that are ubiquitous in urban areas across emerging economies. While these activities are not highly energy-intensive in the sense of large-scale manufacturing, they are central to employment and local economic activity and often rely on electricity-consuming appliances such as refrigeration, lighting, televisions, fans, and small motors. We also study households that use electricity for leisure, comfort, and production of home goods and services. In this setting, electricity quality is economically important because commonly-used appliances are sensitive to sustained under-voltage and voltage fluctuations. These power quality issues impose costs on commercial and residential appliance use: machinery cannot operate at full capacity and protective components can malfunction, damaging appliances.² Even modest voltage improvements could substantially reduce appliance stress and failure rates.

Combining the sensor and survey data, we generate some of the first evidence on the large-scale economic costs of poor voltage quality. We first characterize voltage quality issues and their associated economic costs, and then estimate the causal impact of a quasi-random \$13.9 million investment in electricity grid infrastructure on electricity quality and socioeconomic outcomes. The analyses generate three key findings. First, we document significant voltage problems. Average voltage at baseline – before grid infrastructure investments were completed – is 219V, significantly lower than the targeted nominal level of 230V. Voltage is more

²Elphick et al. (2013) for example state, “an argument can be made that voltage sags are the most costly of all power quality disturbances because of costs associated with lost production” (p. 576).

than 10% (20%) below the nominal level for 5.3 (1.2) hours per day. Customers experience an average of 250 low-voltage ‘spells’ each month, lasting more than 130 hours in total.

Second, these voltage issues are salient to respondents and bring economic costs. Respondents would be willing to pay an additional 10% per month in electricity costs (\$18.5 per year) if their connection had no voltage fluctuations, and 19% more per month if it also had no power outages. One-third of firms report voltage to be an important obstacle to operations. A quarter of respondents own devices to protect appliances against bad voltage, costing them \$56 on average (approximately 15% of both monthly household income and monthly firm revenues in the sample). Despite such investments, 25% of respondents report that at least one appliance was damaged due to bad voltage in the last year, costing an average of \$41 to repair or replace.

Finally, we use a difference-in-differences approach to assess the effects of voltage quality improvements, comparing 76 sites where a new transformer was installed on the local distribution grid to 75 comparable sites without such a transformer investment. We find that the treatment group’s voltage increased from 219V to nearly the targeted 230V, though the control group’s average voltage also rose from 219V to 225V due to other concurrent grid infrastructure investments. The intervention thus increased average voltage by 5.5V in the treatment group relative to the control group, reducing time spent with low-quality voltage by 55 hours per month (44%). Importantly, the voltage quality improvements occurred in a setting that also experiences power *outages* that were not impacted by the new transformer investments.

In the 16 months following the voltage improvements, we find suggestive evidence that treatment reduced voltage-related appliance damage by approximately 20% and ownership of voltage-protective devices by about 8%. At the same time, we find no statistically significant effects on other outcomes, including profitability, labor input, or operating hours among micro-enterprises, or income, employment, and health spending among households. Notably, there is no significant impact on appliance ownership during this period, which could ex-

plain the absence of major effects on broader socioeconomic indicators. Our estimates are sufficiently precise to rule out large short-run effects on firm and household outcomes of the discrete improvement in voltage.

Two interpretations could explain the absence of short-run socioeconomic effects: voltage improvements are economically inconsequential, or households and firms delay capital adjustment until improvements are perceived as sustained. The evidence is more consistent with the latter. Our results provide strong evidence that voltage quality is an important aspect of reliability: thirty-one percent of firms report that voltage fluctuations are an obstacle to operations, and stated preference elicitation suggests that respondents are willing to pay an additional \$3.30 per month (a 19% increase in electricity spending) for electricity with no voltage fluctuations and no outages. They are also willing to pay \$1.50 for a connection with no voltage fluctuations and half their baseline outages, and \$1.80 for a connection with no outages and half their baseline voltage fluctuations.

Furthermore, we observe immediate responses in mechanical outcomes as a result of voltage improvement: respondents report reductions in bad voltage hours, appliance damage falls, and ownership of voltage protection devices decreases. This strongly suggests that voltage improvements were meaningful and salient to consumers. Despite these improvements, economic outcomes that require investment do not change. For instance, we observe no changes in profit, labor input, or operating hours for firms, and no changes in employment or income for households within 16 months after treatment.

This pattern is in line with the absence of capital adjustment. Within 16 months of line bifurcation, we observe no treatment effect on appliance acquisition, which is a direct channel through which voltage improvements can impact economic activity. In our sample, just 37% of respondents acquired any new electric appliance between baseline and endline, and many of these were replacements for existing appliances. Consistent with forward-looking but lagged adjustment, treatment-site respondents are 7.4 percent more likely to report plans to purchase a new appliance in the subsequent year, although this estimate is statistically

imprecise. While our data do not allow us to directly observe the timing of appliance adoption beyond the 16-month horizon, these results are in line with previous evidence that appliance adoption is slow, especially in settings with a long history of unreliable service (e.g., Dinkelman et al., 2024). Inertia in perceptions of power quality may also explain the lack of short-term appliance investment. Respondents in the study area may not trust the persistence of the voltage improvements they observe. In line with this mechanism, the line bifurcation treatment does not reduce customers' willingness to pay for improved electricity connections or perceptions about the likelihood that overall power quality and reliability will improve in the next year.

We find little support for alternative explanations of the null effects. These include 1) a weak first stage, 2) outages that can overshadow voltage improvements, or 3) spillovers that can attenuate our treatment effects. We show that the results do not depend on a variety of baseline characteristics, including baseline outage or voltage levels. Even for respondents with the worst baseline conditions, such as those who owned sensitive appliances with no protective devices and faced the lowest levels of voltage, we show no meaningful effects on economic outcomes. We also find no evidence of spillovers in voltage improvements from treatment to control sites.

One limitation of our study is the composition of the sample, which focuses on households and micro-enterprises rather than larger, more energy-intensive firms. Because these respondents rely less heavily on electricity in production, improvements in power quality may generate smaller or slower economic responses. Indeed, Chaurey and Le (2022) find that the benefits of a large rural electrification program in India are largest in more electricity-intensive industries. At the same time, focusing on micro-enterprises is policy-relevant. Small firms account for a large share of employment and income generation in emerging economies, and in Chaurey and Le (2022) the employment gains from electrification are driven primarily by micro-enterprises. Moreover, evidence from other contexts shows that improvements in power quality can affect household and small-firm behavior. For example, Meeks et al. (2023)

document changes in household appliance purchases following improvements in power quality. In addition, within our sample, we find no differences in economic impacts by measures of baseline electricity reliance (for firms), willingness to pay for electricity improvements, or vulnerability to voltage issues. These findings suggest that while our sample composition may influence the timing or magnitude of responses, it is unlikely to fully explain the absence of short-run economic effects in our setting.

Taken together, our results establish voltage quality as a distinct and economically relevant dimension of electricity service. We document how voltage quality varies at scale in a setting of households and micro-enterprises, how it responds to a realistic distribution network upgrade, and how customers perceive and value these improvements. The intervention substantially improves voltage quality and reduces voltage-related appliance damage and defensive investments. However, we observe no change in appliance ownership over the study horizon and no detectable effects on profits, labor input, or household income within 16 months. This pattern suggests that the economic impacts of voltage improvements operate primarily through capital adjustment rather than immediate changes in production. Because most productive uses of electricity in this setting require durable appliances, the benefits of improved voltage quality may materialize only gradually as firms and households adjust their capital stock.

By establishing these relationships, the paper clarifies the channels through which voltage quality can affect economic activity and the time frame over which behavioral responses may emerge. At the same time, much remains to be learned about the longer-run impacts of voltage improvements. Voltage problems are widespread across emerging economies, which differ in urban structure, baseline outage exposure, and the role of electricity in productivity-enhancing activities. Power quality issues are likely to become more frequent and severe in the absence of infrastructure investments as increases in incomes and average temperatures create more energy demand and add load to local electricity distribution networks (Berkouwer et al., [2026a](#)). The GridWatch measurement infrastructure and empirical framework

developed in this paper provide a platform for evaluating a broad class of power distribution investments and their short-, medium-, and long-run economic effects.

This paper builds on a small set of research papers documenting voltage quality issues in LMICs. Blimpo and Cosgrove-Davies (2019) report that one-third of surveyed enterprises in Tanzania report appliance damage from voltage fluctuations. In a 2018 survey conducted in Cambodia, Ethiopia, Kenya, Myanmar, Nepal, and Niger, 28% of schools and health centers reported that damaged equipment due to voltage fluctuations was a constraint to operations (WB, 2020). Jacome et al. (2019) directly measure voltage levels among 25 households in Tanzania and find that average voltage is often more than 10% below nominal. We use data from the first large-scale collection of customer-level power quality and reliability data to document voltage quality challenges and the impact of a large grid investment program on both a variety of power quality measures and on customer outcomes.

Our paper complements the findings in Meeks et al. (2023), with several important distinctions. Meeks et al. (2023) randomize the installation of smart meters across 20 transformers in the Kyrgyz Republic. The treatment improved voltage quality and led to an increase in electric heater ownership and a rise in electricity consumption during the winter peak period. Two features of their setting differ from ours. First, the smart meter intervention served multiple functions, including automatically disconnecting power when voltage fluctuated beyond acceptable bounds. This mechanism may have provided households with an immediate and salient signal that voltage instability had been mitigated, potentially accelerating behavioral responses relative to the transformer upgrades we study. Second, the primary appliance margin in the Kyrgyz Republic setting is electric resistance heating. Such devices are relatively robust to voltage sags: reduced voltage lowers heat output but does not typically cause permanent damage. In contrast, the appliances prevalent in Ghana – refrigerators, fans, televisions, and small motors – are more sensitive to sustained under-voltage and fluctuations, which can impair performance and generate irreversible damage. These technological differences imply distinct adjustment margins across the two contexts. Specifically,

these differences suggest that voltage improvements may generate immediate consumption responses when they relax seasonal heating constraints, but may operate through slower capital-adjustment margins in environments like ours where voltage primarily affects appliance durability.

This paper also expands our understanding of the economic impacts of infrastructure quality in low- and middle-income countries, and in particular when construction is outsourced to the private sector (Gertler et al., 2022; Hallegatte et al., 2019; Rentschler et al., 2019). Infrastructure quality has historically been difficult to study due to data limitations, as related data are often provided by the state and may therefore not be suitable for independent quality evaluation. We contribute to a literature that collects fine-grained data on publicly-provided infrastructure independently from the public sector (Olken, 2007; Wolfram et al., 2023), and provide the first large-scale estimates of the economic costs of an understudied aspect of modern infrastructure.

2 Voltage quality: importance and measurement challenges

Existing global energy policy focuses overwhelmingly on access and reliability. The World Bank’s Energy Sector Management Assistance Program Multi-Tier Framework quantifies these dimensions in nuanced detail, delineating 12 categories of capacity and 10 categories of availability, with metrics for each category (ESMAP, 2015): for example, the ‘availability’ category contains five tiers for daily availability and five tiers for evening availability.

However, it only defines two coarse tiers of voltage quality: “voltage problems that damage appliances” and “voltage problems do not affect the use of desired appliances” (ESMAP, 2015). Yet even this imprecise definition of voltage quality is difficult to measure. This very coarse categorization of what is a highly complex phenomenon may be driven by limited data availability on voltage. Without granular data, voltage quality cannot be

adequately factored into grid planning or tracked using development indicators.

As a result, the economic implications of poor voltage quality have received little attention. Much of the existing evidence is anecdotal or based on survey reports (Blimpo and Cosgrove-Davies, 2019; Dubey, 2020), with few studies able to collect and analyze detailed measures of customer-level voltage quality. Jacome et al. (2019) measure voltage quality for 25 households in Unguja, Tanzania, and find that customers near the end of the line experience voltage outside the nominal range around half the time. Meeks et al. (2023) analyze records for 20 transformers in Kyrgyzstan and find that they record 2.3 voltage fluctuations per day, with smart meters likely generating improvements.

Utility regulators normally set a target voltage for electricity distribution, and limit the amount of time that voltage, as experienced by customers, can deviate from the nominal voltage level by more than $\pm 10\%$. Appliances are designed to be used on grids that provide voltage in this range. Most of the world’s population – including Ghana – has a nominal voltage of 230V (IEC, 2023). Meeting these requirements is critical to ensuring a high-quality electricity supply, and maintaining secure and stable power systems, which is critical for any modern economic activity (IEA, 2022).

Voltage in low- and middle-income country (LMIC) contexts often falls outside the nominal range. However, a lack of large-scale data has limited research on the broader economic impacts of poor voltage quality. Most utilities and regulators have no way of measuring voltage as experienced by customers.³ This problem is exacerbated in LMICs, where resource constraints prevent utilities from investing in improved technologies: the widespread deployment of smart meters, for example, can be prohibitively expensive (Dutta and Klugman, 2021).

³Many utilities have substation-level monitoring systems, but these only detect outages at high- or medium-voltage transmission levels (while outages emerging at the low-voltage customer distribution level can comprise a large share of power outages Berkouwer et al., 2024), and do not measure customer-level voltage. Transformer-level systems also do not capture customer-level electricity quality.

2.1 Types of voltage quality issues

When voltage falls below the nominal range, appliances often cannot function properly. Lightbulbs will dim or flicker. Some appliances cannot be turned on, particularly if voltage falls to more than 20% below nominal. Some will experience failure of protective components, even as other components continue to function, burning appliances. Voltage spikes – extreme increases often lasting seconds or less – can also cause significant damages to plugged-in appliances. These are rare but sometimes occur as power returns after an outage. Over-voltage spells – modest but longer-lasting increases above the nominal voltage range – are less damaging than under-voltage spells as well as less common. We present data on these phenomena in [Section 3](#).

How voltage fluctuations affect appliances is complex, non-linear, and not well understood. The amount of power used by an appliance (in watts) is a combination of voltage (in volts) and current (in amperes). Counterintuitively, voltage *drops* can cause appliances to burn, either because the drop in voltage prevents protective components of appliances to function properly (causing other components to operate without necessary protections), or because the appliance draws in additional current to compensate for the drop in voltage. These processes furthermore vary significantly across appliances, depending on what electrical components the appliance contains.⁴ A single short but large voltage spike or sag can cause more damage than a more moderate but lengthier under- or over-voltage spell, but a simple average voltage metric will not capture this non-linearity. Fluctuations may also affect appliances differently than under- or over-voltage spells. Unlike reliability, where the System Average Interruption Duration Index (SAIDI) is a globally and industry-wide accepted indicator (Ayaburi et al., 2020; Vugrin et al., 2017), existing indicators for voltage quality do not accurately capture voltage issues common in low-resource settings, instead applying more to high-income settings with only minimal deviations from nominal (IEEE,

⁴This can include, for example, motors, lighting, power electronic devices, or resistive elements (McKenna and Keane, 2016).

2018).

To characterize voltage quality, we define and analyze several metrics: average voltage, time spent outside of certain voltage thresholds, the count of under-voltage spells, the duration of under-voltage spells, and the intensity of voltage spells as measured by minimum voltage reached.

2.2 Causes of poor voltage quality

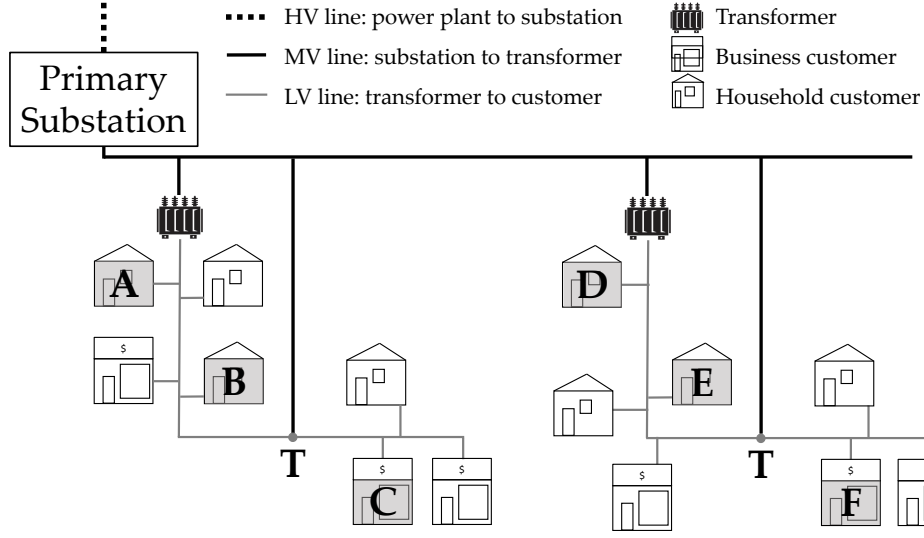
In most countries, high voltage (HV) cables transmit electricity from power stations to primary substations. Medium voltage (MV) cables then transmit this electricity on to distribution transformers, and low voltage (LV) distribution lines then distribute power to residential and commercial customers. [Figure 1](#) presents a schematic.

Transformer load and distance to transformer are key drivers of poor voltage quality. Transformer load is the aggregate electricity demanded by all customers connected to a transformer. As load increases, the transformer’s output voltage drops, causing voltage to drop. Load variability therefore also increases customer voltage variability. If the transformer is overloaded – that is, when load exceeds transformer capacity – output voltage can drop below the target voltage range. Frequent overloading of transformers can also lead to failures causing local outages (Carranza and Meeks, [2021](#); Glover et al., [2012](#)).

Distance to the nearest transformer worsens voltage due to impedance in LV lines and due to the increased load between the transformer and the customer (Bailey et al., [2023](#); Berkouwer et al., [2026a](#); Jacome et al., [2019](#); Wolfram et al., [2023](#)). The electricity grid intervention that we study (which we discuss in detail in [Section 5](#)) adds (or ‘injects’) new transformers to the grid, reducing the average load on existing transformers and also reducing the distance between customers and their nearest transformer.

[Figure 1](#) visualizes these dynamics in an example grid. Customers A and D experience similar voltage because both are near the transformer, and their transformers have a similar load. E might experience worse voltage than D because they are farther from their trans-

Figure 1: Schematic of an electricity distribution network



Schematic of a radial electricity distribution network. Transformers step down voltage and distribute it to household and firm customers along low voltage lines.

former, but better voltage than B because there are fewer customers between E and the transformer than between B and the transformer. Adding two new transformers at T should improve electricity quality for C and F, and likely for B and E, by reducing the distance to their nearest transformer. It might also improve power for A and D by reducing the load on their nearest transformer.

Baseline customer socio-economic characteristics are not strongly correlated with power quality: an index of customer wealth (proxied by structure quality, appliance ownership, and education) is not significantly associated with either bad voltage or outage hours ([Table B1](#)). This suggests that customers in our sample are not sorting on attributes. Distance to the nearest transformer is the strongest predictor of voltage quality: being 100 meters farther away is associated with 22 additional hours with voltage $>10\%$ below nominal per month ([Figure A1](#) reveals a non-linear relationship). Outage hours also increase with distance to the nearest transformer.

Power outages can also impact voltage quality, through transient spikes that occur at the inception and restoration of outages, causing significant damage to appliances. These

phenomena are of very short duration and will not be captured in the voltage measurements we consider here, and are also not expected to lessen as a result of transformer injections. Other unobserved aspects of grid quality – such as phases, household connection quality, and informal connections – also likely affect voltage quality and socio-economic outcomes.

3 Measuring power quality in Ghana

Ghana achieved high levels of electricity access earlier than most sub-Saharan African countries, with 64% of households connected in 2010 compared to the regional average of 33% (WB, 2010). From 2012–2016, Ghana experienced a severe power crisis with periods of rolling blackouts in the face of power shortages. This crisis has been covered in the media (The Guardian, 2015; Al Jazeera, 2016; New York Times, 2016; BBC, 2016) and in academic research (Abeberese, 2019; Abeberese et al., 2021; Briggs, 2021; Hardy and McCasland, 2019; Yakubu et al., 2018).⁵ Access and reliability have improved significantly in Ghana in recent years, with household electricity access now 86% (SE4All, 2022; Kumi, 2020) and outage duration down to 30 minutes per day per our data. Less attention has been given to the quality of the electricity when it is flowing, even though the problem is salient for many Ghanaians. While Google trends indicate on average more searches of terms related to outages, searches for ‘low current’ are often of similar magnitude (Figure A2).

We collaborated with engineers to deploy 627 GridWatch devices collecting minute-by-minute power quality and outage data with customers residing across the Accra metropolitan region starting in 2018 (Figure A3 shows a picture of the device).⁶ Each device is plugged

⁵At the height of the crisis, consumers faced 24-hour power cuts every 36-hour period (Mensah, 2018; Prempeh, 2020). Using data on outages at the electricity feeder (MV) level from the electricity utility in Accra, we find a peak of over 250 outage hours per month on average in July 2015, and over 100 outage hours per month for all of 2015.

⁶The total number of devices deployed across the Greater Accra Region is 1,124, of which 627 were deployed in the study sites for this paper. Devices were deployed to customers residing near the locations of potential grid infrastructure investments, which we describe further in Section 5. The devices do not capture very short sags or swells that last only a few cycles or seconds. This would require high-frequency, continuous waveform monitoring, such as those provided by significantly more expensive power quality monitors. However, line bifurcation likely will not reduce these extreme events as these are mitigated by

in with either a household or firm connected to the electricity grid.⁷ This generates 337 million data points on voltage as experienced by households and firms. To the best of our knowledge, this is the first large-scale collection of customer-level outage and voltage data in any low- or middle-income country.⁸

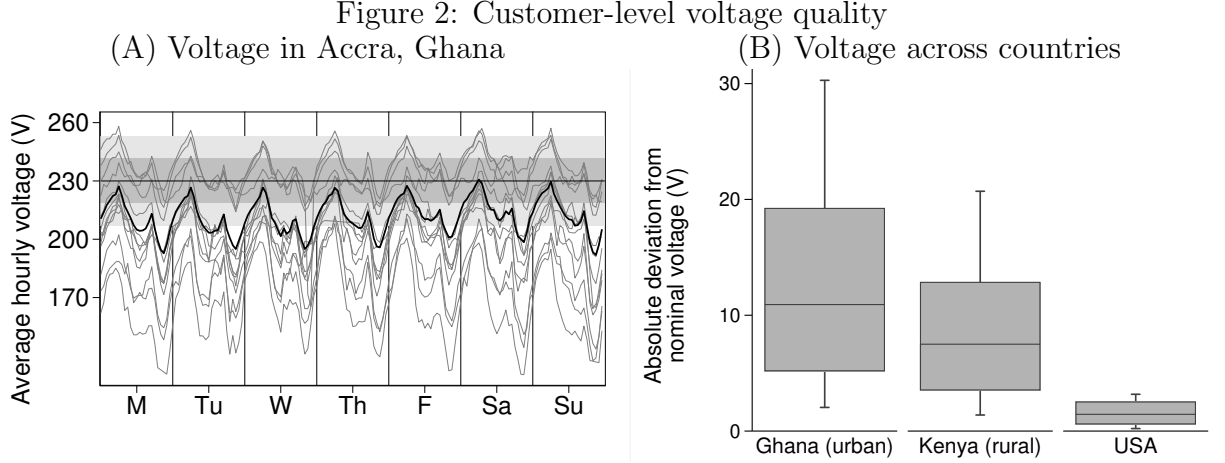
Outages are identified by detecting spatial and temporal clusters of outage signals from multiple devices. Thus, if one individual unplugs the device or lets their pre-paid meter run out of tokens, this will not be recorded as a power outage because there are no concurrent signals from nearby devices. This detection algorithm also identifies whether an outage originates in the low-voltage, medium-voltage, or high-voltage part of the network: if several devices connected to a single transformer report outage signals within seconds of each other, this outage likely originated in the low-voltage part of the network. On the other hand, if devices that are connected across all transformers connected to a single primary substation report outage signals this is classified as an outage originating in the high-voltage part of the network.

Most modern electronic equipment is rated for input voltage between 100–240V (Elphick et al., 2013). This means that even if voltage falls to 50% (115V for a 230V nominal system), voltage is still within the operating range. However, in Ghana, few appliances are designed to function at low voltage levels. Ghana’s nominal voltage is 230V, and appliances in Ghana are most often rated for 220–240V, making them more vulnerable to moderate voltage fluctuations than appliances used in higher-income contexts which allow a larger input voltage range. Our firm and household surveys in Accra find that wide voltage ratings (such as 100–240V) for major appliances are rare, though ranges listed on devices may be

other investments (such as fuses and switches). Klugman et al. (2019) and Klugman et al. (2021) provide more information on the technology and the deployment. Berkouwer et al. (2026a) combine the Ghana GridWatch data with data from GridWatch deployments in three other African countries to analyze power quality and reliability and their implications for Africa’s energy transition.

⁷Each participant receives financial compensation for each month they keep the device plugged in. [Section 5](#) presents more information on site and respondent selection.

⁸Jacome et al. (2019) measure customer-level voltage levels among 25 households in rural Tanzania. Meeks et al. (2023) deploy smart meters which record voltage events to households connected to 20 transformers in Kyrgyzstan, but these do not record high-frequency voltage levels and it is unclear what types of events are recorded.



Panel A: Voltage measurements for 20 randomly selected participants for an arbitrarily chosen week in April 2020. The bold line displays the average over the entire sample. The shaded horizontal bands indicate $\pm 5\%$ and $\pm 10\%$ outside nominal voltage (230V). *Panel B:* 10th, 25th, 50th, 75th, and 90th percentiles of device-level voltage. Kenya data from Wolfram et al. (2023). U.S. data from Pecan Street (2018).

conservative. Voltage stabilizers, typically rated for 140-260V or 105-280V in Ghana, can be used to modify incoming voltage.

Ghana’s public utilities regulator specifies that electric utilities must provide electricity with sustained voltages between $\pm 10\%$ of the nominal level of 230V, allowing for larger deviations only for very short duration (PURC, 2005). Actual voltage deviates substantially from these targets. Panel A of Figure 2 displays data for 20 devices. Several patterns are worth highlighting. First, there is significant heterogeneity in average voltage across customers, which as discussed in Subsection 2.2 may reflect differences in distance to the nearest transformer and in transformer load (Figure A1). Second, customers often experience fluctuations outside the recommended range. Third, voltage is consistently worst between 7–10pm, when load is likely highest. Fourth, nearly all deviations outside the nominal range constitute voltage *drops*.

Panel B of Figure 2 compares Ghana’s average absolute deviation from nominal voltage with data from Kenya (Wolfram et al., 2023) and the U.S. (Pecan Street, 2018). Voltage in the U.S. is within 3% of nominal voltage 95% of the time. In Kenya and Ghana, median voltage deviates around 10V from nominal, with Ghana often experiencing even more significant deviations.

Table 1 presents statistics for several electricity indicators designed to give a comprehensive picture of customer-level voltage quality, over the period from March 2019–November 2020 (before local grid improvements). Average voltage was 219V, outside the voltage rating range for most appliances in Ghana (220–240V). The median absolute deviation from nominal voltage was 12V, driven primarily by low-voltage spells. In a given hour, voltage was on average 10–20% below nominal 17% of the time and more than 20% below nominal 5% of the time. During peak load periods, the fraction of time more than 10% below nominal exceeds 30%. Voltage quality issues are significantly more common than power outages, which occur 3% of the time (14 hours per month).

Panel B presents data at the monthly level, which enables a characterization of the frequency and duration of sustained under-voltage spells. Consider low voltage events in which the minimum voltage fell to 184–200V, which is outside the voltage ratings for most appliances in Ghana. On average, customers experienced 32 such spells per month, lasting a total of 31 hours. For more severe spells where the minimum voltage was less than 184V (more than 20% below nominal), customers experienced on average 11 such spells per month, lasting a total of 84 hours. More extreme under-voltage spells are concentrated among customers with the worst power quality: the median customer experienced 10 hours of outages and 16 hours of low voltage per month.

4 Self-reported economic costs of poor voltage quality

Voltage quality can affect economic productivity and well-being in four main ways. First, poor voltage can restrict the productive use and utility of electric appliances. Second, it can damage appliances and require spending on repairs or replacements. Third, it can require investments in devices designed to protect against voltage fluctuations (such as voltage stabilizers) or in backup energy sources. Finally, these mechanisms may lower long-term appliances investment by lowering their expected value.

Table 1: Summary statistics for baseline measures of power quality
(A) Hourly data

	Mean	SD	Min	25 th	50 th	75 th	Max
Mean voltage during hour	218.92	21.51	23	209	222	233	418
Mean voltage during hour (7-10pm)	209.69	23.63	23	197	213	226	409
Mean absolute deviation from nominal	16.88	17.38	0	6	12	22	207
Mean absolute deviation from nominal (7-10pm)	23.28	20.74	0	8	17	33	207
Any voltage >20% below nominal	0.09	0.28	0	0	0	0	1
Any voltage >20% below nominal (7-10pm)	0.19	0.39	0	0	0	0	1
Share of hour voltage >20% above nominal	0.00	0.01	0	0	0	0	1
Share of hour voltage 10-20% above nominal	0.02	0.11	0	0	0	0	1
Share of hour voltage 10-20% below nominal	0.17	0.33	0	0	0	0	1
Share of hour voltage >20% below nominal	0.05	0.21	0	0	0	0	1
Share of hour with no power (outage)	0.03	0.15	0	0	0	0	1
SD voltage during hour	2.59	3.02	0	1	2	3	159

(B) Monthly data

	Mean	SD	Min	25 th	50 th	75 th	Max
Hours with no power (outages)	14.24	14.86	0	4	10	20	146
Number of spells with min voltage >200	206.96	243.81	0	1	109	342	922
Number of spells with min voltage btwn 184-200	31.74	48.30	0	0	10	43	224
Number of spells with min voltage <184	10.77	16.81	0	0	2	16	97
Total duration of spells with min voltage >200	15.43	17.89	0	0	8	27	70
Total duration of spells with min voltage btwn 184-200	31.10	42.76	0	0	8	52	181
Total duration of spells with min voltage <184	84.15	160.12	0	0	0	79	641

Summary statistics for power quality measured by 441 devices across 138 sites between March 2019 and November 2020 (before transformer injections). Data for this period are not available for the other 186 devices deployed later in the study sites. Outages are identified at the site level. *Panel A*: 2,872,508 device-hour observations, each computed from thirty 2-minute observations. *Panel B*: 6,871 device-month observations, summing either hourly values by device within each month (rows 1) or across individual low-voltage spells recorded by device within each month (rows 2–7). Low-voltage spells are periods with voltage <207V for at least two minutes.

To examine the economic costs of poor voltage quality, we survey 2,001 electricity grid customers – 997 households and 1,004 firms – across 151 distinct study sites in Accra where GridWatch devices were deployed. As we discuss extensively in [Section 5](#), these sites are the locations of potential new transformers. To ensure data collected by the devices reflected power quality experienced by survey respondents, respondents and devices were located very close together (see [Figure A6](#) for an example map).⁹ Per data from the Ghana Statistical Ser-

⁹To avoid survey fatigue, there is no overlap between participants with devices and survey respondents.

vice, sample respondents are largely representative of households in the Accra Metropolitan Area (Table 4). Because the new transformer investments largely targeted mixed residential and business areas, firms in the survey sample are primarily micro-enterprises with one or two employees including the owner or manager, and consequently employees, revenues, and profits are lower than the Accra median. The most common firm activities are small retail operations (44%), personal care services such as hair and nail care (16%), manufacture and repair of clothing (15%), and food and beverage services (5%).

4.1 Customer self-reports of reliability issues and associated costs

The baseline surveys that were conducted in March–April 2021 reveal that respondents face poor power quality. Panel A of Table 2 shows that respondents report experiencing 39 hours of outages and 48 hours of bad voltage in the past month. In addition, 26% of respondents reported having voltage-related damage in the past 12 months, as a result of which they spent \$41 to repair or replace broken appliances (around 2.5 months of average electricity spending, or around 12% of monthly household income and firm revenue).¹⁰ As overheating or burning appliances can cause human harm (or broader damage in the case of fire), we also ask respondents about injury caused by power quality issues in the last 12 months. This question was only asked at endline, where 11 respondents (0.7% of the endline sample) report voltage-related injuries and 6 of these report that these resulted in medical costs and/or missed work.¹¹ Though these events are rare the potential costs can be large, contributing to the overall costs of poor power quality.

To protect against these damages, customers can purchase equipment that protects appliances from bad voltage: 25% of respondents have at least one voltage protective device, with an average estimated value of \$56.¹² Power issues can also reduce household utility or

¹⁰All monetary values are in 2022 USD, converted from Ghana Cedis using the average 2022 exchange rate and adjusting baseline values for local inflation between 2021 and 2022.

¹¹As we discuss in Subsection 5.4, voltage quality improved between the baseline and endline surveys, so the risk of such injuries may have been higher at baseline.

¹²These include general purpose voltage stabilizers (15% of customers) as well as more specialized devices

Table 2: Baseline household and firm electricity quality issues

	Mean	SD	Min	25 th	50 th	75 th	Max	N
<i>Panel (A) Experience with outages and voltage</i>								
Reported number of outages in past month	6.86	5.50	0	4	6	8	90	2001
Reported total outage hours in past month	39.15	47.71	0	12	24	48	300	2001
Reported hours of bad voltage in past month	47.72	100.34	0	0	15	60	720	1988
<i>Panel (B) Economic impacts</i>								
Any voltage-related damage, last 12 months (=1)	0.26	0.44	0	0	0	1	1	2001
Amt. spent on burnt/broken apps in past year (\$)	41.43	129.49	0	0	11	40	2389	511
Any reliability protective device owned (=1)	0.25	0.43	0	0	0	0	1	2001
Value of protective devices owned (\$)	55.71	110.43	2	14	24	53	1290	261
Any alt. energy source used in last month (=1)	0.05	0.21	0	0	0	0	1	2001
Outages are obstacle to business (=1)	0.92	0.28	0	1	1	1	1	975
Voltage fluctuations are obstacle to business (=1)	0.31	0.46	0	0	0	1	1	975
<i>Panel (C) WTP for improved service quality</i>								
WTP for perfect reliability and quality (\$)	3.25	5.01	0	0	2	5	72	2001
WTP for perfect voltage and half outage hours (\$)	1.54	3.14	0	0	0	2	56	1964
WTP for no outages and half bad voltage hours (\$)	1.80	4.10	0	0	0	2	48	512
Share of govt. investment to reducing outages	0.17	0.12	0	0	0	0	1	1061
Share of govt. investment to improving voltage	0.15	0.13	0	0	0	0	1	1061

Spending on burnt appliances is among those with voltage-related damage. Value of voltage protective devices is among those with any such devices. Reliability protective devices include voltage stabilizers, fridge guards, and TV guards. Alternative energy sources include generators, solar panels, and wet cell batteries. Stated WTP values are monthly amounts for a hypothetical electricity connection with particular characteristics. Only a subset of respondents in the baseline survey were asked their WTP for reduced but not eliminated bad voltage hours. Data on share of government investment comes from the endline survey (not asked in the baseline survey) and omits respondents that equally allocated the hypothetical investment across all 5 areas. All statistics pool household and business customers but are broadly similar when considering each separately.

firm revenues by constraining appliance use. Thirty-one percent of firms report that voltage fluctuations are an obstacle to firm operations. While this is much lower than the 92% reporting that outages are an obstacle, it is still a significant share considering the more subtle ways that voltage may affect business operations compared to the salience of outages. Just 5% of respondents have an alternative energy source (generators, solar panels, or wet cell battery): most have no alternative when grid electricity service is poor.

Would customers prefer a utility to invest in reliability or voltage improvements? We
 such as fridge guards (11%) and TV guards (4%).

use a standard set of stated preference questions to measure willingness to pay (WTP) for improvements in the reliability and quality of their electricity connection. We use a binary search to determine the maximum increase in monthly electricity costs the respondent is willing to pay for an improved connection (beyond what they currently spend), first asking about a randomly chosen price and then iteratively asking about either higher or lower prices based on the prior response.¹³ Panel C of [Table 2](#) shows that respondents are willing to pay on average an additional \$3.3 per month (a 19% increase in electricity spending) for access to an electricity connection with no voltage fluctuations or outages. This includes around 30% of respondents with a WTP of \$0. WTP for a connection with no voltage fluctuations and half the respondent’s baseline monthly outages and WTP for a connection with no outages and half the baseline level of voltage fluctuations are similar – \$1.5 and \$1.8 per month, respectively, representing around 10% of monthly electricity spending ([Figure A4](#) presents the full distributions). WTP does not appear correlated with appliance or lightbulb type ownership ([Figure A5](#)).

These WTP values are hypothetical and could be biased upward, particularly as they are asked after survey modules where respondents report on their electricity spending, appliance ownership, and perceptions of electricity reliability and quality. Respondents may also want to express high WTP in the hope of influencing investment policy, though we do not specify how the results will be used. To help calibrate responses, we present information on what respondents reported they currently pay for electricity and the power issues they face in the framing of the WTP exercises.

Hypothetical bias could affect the interpretation of the magnitudes of the WTP responses, though the comparison of WTP for different improved electricity connections should not be affected. To check whether stated WTP magnitudes are realistic, we can compare these values against reported voltage-related costs. On average, respondents state being willing to pay \$18.5 per year for a connection with perfect voltage and half of their current outage

¹³This methodology has been used in Ghana and elsewhere in Africa – see for example Abdullah and Jeanty (2011), Berkouwer et al. (2022), Deutschmann et al. (2021), and Sievert and Steinbuks (2020).

hours. Average spending on dealing with appliances damaged by voltage issues is \$10.8 per year and the average value of protective devices owned is \$13.9, though these devices do not need to be purchased every year. A connection with perfect voltage quality would eliminate all voltage-related damage and any need for protective devices while also providing benefits from improved appliance usability and decreased risk of human harm from large voltage fluctuations. This comparison suggests that the stated WTP values are not unrealistically biased upward.

Another potential concern is whether respondents can accurately disentangle outages and voltage fluctuations as distinct power supply issues. In our context, several factors suggest that respondents do indeed make this distinction. First, 25% of households own voltage-specific protective devices. This revealed preference for protection indicates a clear understanding of voltage as a specific threat to appliances. Second, the WTP survey module was preceded by detailed modules on separate outage and voltage experiences, priming respondents to consider them as independent problems. Respondents are asked follow up questions about when and how bad voltage affects their appliances, the types of appliances most affected, and the nature of damages and health impacts, and demonstrated a clear awareness of voltage-specific appliance issues. Finally, the correlation between self-reported outage and bad voltage hours is modest ($r = 0.32$) and similar to the correlation between measured outages hours and bad voltage hours ($r = 0.24$), suggesting that for most respondents these problems occur at different times and for different reasons. While we cannot rule out some degree of conflation, these facts suggest that respondents perceive outages and bad voltage as distinct economic burdens.

We also ask respondents how they would prioritize the allocation of hypothetical government funds across five public spending categories: reduced power outages, improved voltage quality, improved schools, reduced traffic congestion, and improved access to piped water. Around one-third of respondents report that they would evenly split the allocation across the five areas. Excluding these, respondents would on average allocate 15% of the funds

to improving voltage quality, similar to the mean amount allocated to reducing outages (17%).¹⁴ Improving schools (29% of the funds) and access to piped water (26%) are the relative priorities.

Taken together, the results indicate that in addition to valuing outage reductions, respondents place significant value on the distinct power quality problem of voltage. Ghana’s challenges with electricity reliability – and the associated economic costs – are well-publicized (The Guardian, 2015; Al Jazeera, 2016; New York Times, 2016; BBC, 2016). While voltage quality concerns have seen significantly less public discussion, they may also cause significant economic damages and warrant policy intervention.

4.2 Correlates of customer-level power quality and reliability

To examine how voltage quality correlates with respondent outcomes, we match each survey respondent with their nearest device. Since respondents and devices were deployed in clusters (‘sites’), respondents and devices are generally located close together and often connected to the same or neighboring segments of the LV network (Figure A6). The median distance between a respondent and their matched device is 88 meters, and 80% of the respondents are within 157 meters of their matched device (Figure A7). The distributions of baseline distances to the nearest existing transformer are nearly identical for survey respondents and matched devices (Figure C4).

Such spatial proximity means power quality measured by the devices is a reasonable proxy for power quality experienced by survey respondents. The correlation in hourly voltage data between pairs of devices decreases with distance, but devices that are located within 100 meters of each other have an average correlation coefficient of 0.7 (Figure C1). There is a strong correlation between self-reported outage hours with device-measured outage hours (Panel A

¹⁴The question on allocation of government funds follows the willingness to pay for improved electricity questions as well as modules on appliance ownership and damages, outage and voltage experiences, and alternative energy use. It is possible that respondents have been primed to focus on electricity issues which could bias the share of funds allocated to these services upward. However, it should not bias funds allocated to improved voltage relative to reduced outages as respondents will have been similarly primed to think about both types of issues.

of Figure A8). The correlation between reported hours of bad voltage and device-measured hours when voltage was $>20\%$ below nominal is somewhat weaker, with respondents underestimating recent hours of bad voltage in situations where power quality is very bad and overestimating them when power is better (Panel B of Figure A8).

The weaker correlation for voltage may have several explanations. Individuals may only be able to detect voltage problems when voltage drops farther below the threshold of 20% below nominal voltage we use to proxy ‘bad’ voltage with the device data. They may also not observe voltage issues during periods they are not actively using appliances that are sensitive to voltage.¹⁵

Since distance to the nearest transformer is a key determinant of customer voltage quality, we create an adjusted voltage measurement that accounts for the difference in distance to transformer between the respondent and their matched device and thus generates a more accurate measure of respondent power quality.¹⁶ This adjusted voltage quality measure is more accurately correlated with self-reported power quality (Panel C of Figure A8), without meaningfully affecting standard errors (compare Table 3 and Table C10).

Table 3 shows that better voltage quality over the previous year is correlated with lower probability of appliance damage and associated expenses, and with lower WTP for improved electricity connections. We also find a marginally significant relationship between firm revenue and voltage quality: firms with 11V higher average voltage (the difference between average voltage of 219V and nominal voltage of 230V) earn on average \$20.6 more per month. This could reflect either firm sorting on voltage quality or a causal effect of power quality on revenues. However, we find no relationship between bad voltage quality and appliance

¹⁵Flickering of lights and appliances not turning on are key indicators of bad voltage, but they only occur when voltage drops significantly beyond 10% below nominal voltage and will be most visible at night. In addition, low voltage may be more noticeable with incandescent lightbulbs than with CFLs or LEDs. Only 3% of respondents own incandescent bulbs, while 28% own CFLs and 82% own LEDs (enumerators showed respondents pictures of each lightbulb type to ensure these were coded correctly). This is likely a result of concerted efforts by the utility to switch customers to more energy efficient technologies in order to avert excess load that drove power outages during the Dumsor crisis (Ghana Energy Commission, 2009).

¹⁶Specifically, we estimate the overall relationship between distance to the nearest transformer and voltage quality, storing the residuals; project this relationship onto respondent distance; and then re-apply the original residuals.

Table 3: Correlations between voltage quality and selected outcomes

	Full N	Mean (SD)	Adjusted average voltage	Adjusted hours voltage 20% below nominal
Voltage damage and protection index	3057	-0.02 (0.97)	-0.003* (0.002)	0.001** (0.001)
Amt. spent on burnt/broken apps in past year (USD)	2989	6.56 (22.94)	-0.083*** (0.029)	0.017** (0.008)
WTP for perfect reliability and quality (USD)	3057	2.50 (3.46)	-0.019*** (0.005)	0.003*** (0.001)
Total no. of appliances owned	3057	8.48 (5.38)	0.003 (0.006)	0.000 (0.001)
Last month business revenue (USD)	1245	392.85 (574.60)	1.873** (0.901)	-0.355* (0.213)

This table shows estimates from ten regressions with two voltage measurements as the explanatory variables and each row a different outcome variables, pooling firm and household respondents. Voltage measures are for the year preceding the survey date, adjusted to account for differences between the distance to the nearest transformer for the respondent and their matched device. Regressions include socioeconomic controls and district fixed effects. Standard errors are clustered at the site level. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$ [Table B3](#) presents additional outcomes. [Table C10](#) shows the same using raw matched device data: the estimates and standard errors are very similar.

ownership or other outcomes such as protective device ownership, use of non-grid energy sources, electricity spending, household income, or business profits ([Table B3](#)). Appliance ownership among respondents may be too low for sorting on voltage to be worthwhile, or customers may face difficulties in observing local differences in power quality.

5 Identifying the causal impact of grid investments

In 2014, the Millennium Challenge Corporation (MCC) signed the Ghana Power Compact to disburse \$316 million in funding towards electricity network improvements in Ghana (MCC, 2014),¹⁷ Much of the funding flowed to the Electricity Company of Ghana (ECG) through the Financial and Operational Turnaround Project (EFOT).¹⁸ MCC assumed that the EFOT would generate a reduction in annual power outages of 50% and estimated that this reduction

¹⁷The original amount was \$498 million but this was reduced to \$316 million in 2019 (MCC, 2022).

¹⁸ECG is the sole grid utility in the study area, and provides electricity for most of southern Ghana. Average total annual investment by ECG in ‘payments to acquire property, plant, and equipment and towards capital work in progress’ from 2019-2024, excluding the Compact, was \$392 million (ECG, 2024).

– together with additional grid improvements – would generate an economic rate of return of 19% for the compact as a whole (MCC, 2024).

5.1 Low-voltage line bifurcation

Around 4% (\$13.9 million) of the aggregate investment was spent on “low-voltage (LV) line bifurcation” in three districts of Accra: Achimota, Dansoman, and Kaneshie.¹⁹ We estimate the effects of this investment on power quality and economic outcomes using a difference-in-differences strategy.

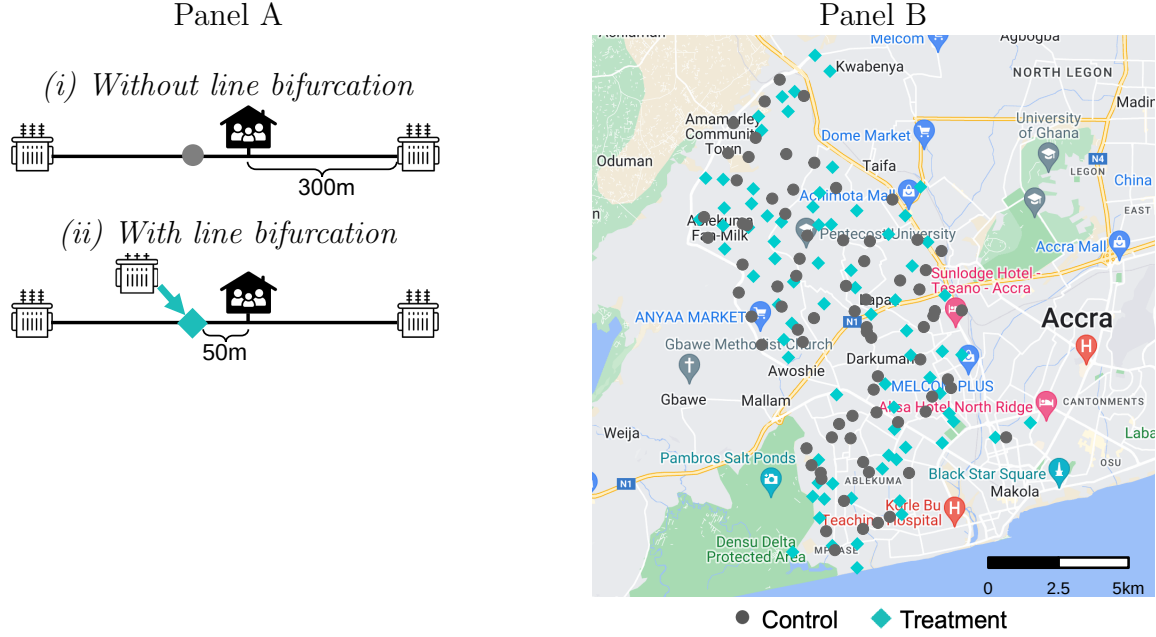
Line bifurcation involves adding a new transformer to the LV network with the goal of reducing average transformer loads as well as “to reduce the length of the low voltage circuits to ensure they do not exceed a length that affects the quality of service and a technical loss threshold” (MCC, 2014). Panel A of Figure 3 provides an illustration. As discussed in Subsection 2.2, the reduction in distance and in transformer load should also directly improve voltage quality, in particular for customers whose distance to the nearest transformer decreases the most.²⁰

An MCC contractor selected the new transformer locations, targeting segments on the grid that were approximately 200 to 300 meters from the nearest existing transformer. Other than this distance criterion, the contractor had very limited local data to inform location decisions. They did not have access to any type of socioeconomic or demographic characteristics, or utility data on things like the number of metered connections, bill payment rates, or electricity demand (sub-district electricity data in general is largely unavailable – the utility

¹⁹The full intervention was described by the MCC as, “Technical losses on the distribution lines have an inverse exponential relationship with the voltage—that is, the higher the voltage the lower the losses [...] When LV lines are long (such as in Accra) and heavily loaded, their thermal losses become a large portion of the technical losses on the system. Moreover the voltage level at the end of the long LV network drops below allowable standards for quality of service to the customer. The purpose of this project is to reduce the length of the 440 volt (LV) circuits (segmenting a large circuits into multiple smaller ones) to ensure that the 440 volt trunk lines do not exceed a length that affects the quality of service to the customer and a technical loss target from the KPIs.” (MCC, 2024)

²⁰In the long term, customers may respond to improved electricity quality by increasing usage, which would worsen voltage quality. This should be weighed against the reduction in transformer load in the short term.

Figure 3: Line bifurcation control and treatment sites across Accra



Panel A presents a schematic of control and treatment sites. Without line bifurcation, the customer is 300 meters from their nearest transformer. With line bifurcation, the distance to the nearest transformer for this customer drops to only 50m. Panel B presents a map of control and treatment sites across Accra, Ghana.

only prepares district-wide data). The one exception to this is that the contractor obtained analog readings of transformer-level load measuring the highest instantaneous load experienced at a transformer since the last reading. However, these must be reset manually and are not reset at a fixed schedule, and are therefore a crude and noisy measure of load. Extensive discussions with representatives from MCC and the contractor confirmed that, conditional on distance to the nearest transformer, line bifurcation treatment sites were selected without obvious regard for the outcomes we study.

This decision-making process for the placement of new transformers created quasi-random variation in power improvements, which we exploit for our analysis. The contractor selected 76 locations for transformer injection ('treatment sites'), but many more sites could have been selected. Using spatial data covering the entire electricity network in Accra, our research team identified other segments of the LV grid that were between 200 and 300 meters from both any existing transformers and any treatment locations – thus following the main criterion for treatment site selection – and then randomly selected 75 locations from this set

(‘control sites’). The 76 treatment sites and 75 control sites, shown in the map in Panel B of [Figure 3](#), comprise the 151 sites of our study sample.

We then use maps of the distribution network to define the boundaries for data collection in each site ([Figure A6](#) presents an example). We first identify segments of LV lines that are <200 meters from the new or placebo transformer location and >300 meters from any existing transformers. Customers within 25 meters of these LV segments are those whose electricity service would likely be affected by a transformer injection. Defining these boundaries also reduces the likelihood of spillover power improvements in control sites from nearby transformer injections.

5.2 Data

We began collecting power quality and reliability data in March 2018 with all 151 sites covered by March 2019 (see Klugman et al. (2019) for more detail on the deployment methodology). We focus on data collected between March 2019 and April 2023, encompassing the transformer construction period which lasted from October 2020 to March 2021 ([Figure A9](#) presents a timeline).

Baseline surveys with 6-7 firms and 6-7 households in each site were conducted in March–April 2021 and endline surveys in July–September 2022.²¹ As discussed in [Subsection 4.2](#), there is no overlap between survey respondents and respondents that received a GridWatch device.

Of 2,001 respondents surveyed at baseline, 1,575 were surveyed at the endline one year later. Attrited respondents are similar to non-attrited respondents along most socioeconomic characteristics, though they differ along certain variables commonly associated with attri-

²¹Due to COVID-19-related delays, baseline surveys were conducted while line bifurcation construction activities were being completed. However, voltage quality did not improve significantly until April 2021 ([Figure 6](#)). In addition, the short period between construction and baseline surveys is likely to have been too short for households or firms to notice any sustained improvement, let alone act on this improvement and have it reflected in socioeconomic outcomes – most of which are measured over the month or year prior to the survey date. In support of this, we find no baseline differences in respondent outcomes by treatment status ([Table B5](#)).

Table 4: Baseline balance

	Control N	Mean	Treat N	Difference	<i>p</i> -val	Accra Mean
<i>(a) All respondents</i>						
Completed secondary education (=1)	772	0.50	803	-0.00	0.983	
Owns premises (=1)	772	0.37	803	-0.02	0.368	0.59 ^a
Last month electricity spending (USD)	763	17.59	796	1.78	0.026	15.34 ^a
Any television (TV) at location (=1)	772	0.68	803	-0.05	0.040	0.85 ^b
Any fridge at location (=1)	772	0.63	803	0.02	0.434	0.62 ^b
Any reliability protective device owned (=1)	772	0.25	803	-0.02	0.461	
Reported number of outages in past month	772	6.98	803	0.35	0.139	
Reported total outage hours in past month	772	38.61	803	-1.47	0.541	
Reported avg. hours per day with bad voltage	769	1.44	797	-0.29	0.062	
Appliance damaged by voltage in past year (=1)	772	0.25	803	-0.04	0.056	
Spending on burnt/broken apps in past year (USD)	768	9.18	794	-0.25	0.877	
<i>(b) Households only</i>						
Last month household income (USD)	347	330.63	367	-7.06	0.846	328 ^a
Adults with paid jobs	363	0.64	383	-0.03	0.265	
<i>(c) Businesses only</i>						
Number of workers	409	1.99	420	0.04	0.790	7 ^c
Last month business revenue (USD)	343	397.94	380	-0.54	0.990	7188 ^c
Last month business profit (USD)	310	102.40	336	8.18	0.457	1851 ^c
Usual business open hours	409	12.16	420	0.19	0.268	

This table shows means in the baseline period for survey respondents, pooling businesses and households, and tests for significance of the differences in means by line bifurcation treatment status. Summary statistics for the population of households in Accra are taken from Ghana Statistical Service data from the 2017 Ghana Living Standards Survey (^a) or the 2015 Labor Force Survey (^b). Summary statistics for the population of businesses in Accra are taken from Ghana Statistical Service data from the 2015 Integrated Business Establishment Survey II (^c). [Table B2](#) presents additional outcomes, joint F-test results, and additional detail. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

tion such as age, household size, and rental status ([Table C1](#)). Importantly, there are no socioeconomic differences between attrited respondents in treatment and control locations with the exception that respondents that attrited from treatment locations were less likely to own their premises ([Table C2](#)).

To verify compliance with planned transformer injections, we use construction progress reports submitted by the private contractor, tracking each site. We also conducted site visits between November 2020 and October 2021 to confirm the presence (absence) of new transformers at treatment (control) sites. [Subsection 5.5](#) presents robustness to heterogeneity in construction completion.

5.3 Baseline balance between treatment and control sites

To lend support to the quasi-random nature of the assignment mechanism, we conduct a battery of tests examining baseline differences between control and treatment sites. [Table 4](#) shows that respondents' socio-economic characteristics at baseline are balanced across treatment and control sites ([Table B2](#) presents additional outcomes).²² In line with the site selection procedure described in [Subsection 5.1](#), the distributions of distance between each survey respondent and the nearest existing transformer prior to line bifurcation are statistically indistinguishable across control and treatment sites (Panel A of [Figure A12](#)).²³

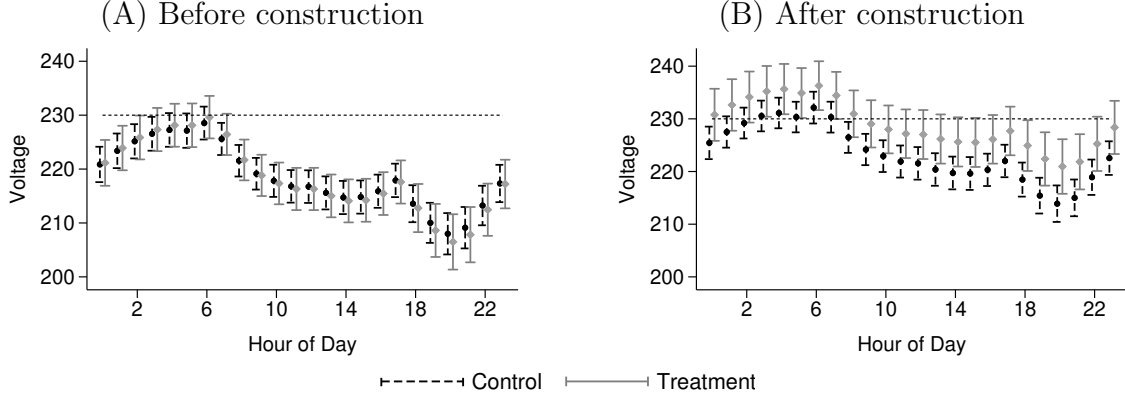
Levels and pre-trends in outages and voltage by site treatment status are statistically indistinguishable before the line bifurcation intervention across a host of outcomes. Panel A of [Figure 4](#) shows that treatment and control sites have indistinguishable voltage patterns across all hours of the day in the pre-period. [Figure 6](#) shows parallel pre-trends for average voltage and outages in the 18 months leading up to the intervention ([Figure A10](#) shows parallel pre-trends for absolute voltage deviation, standard deviation, hours of spells >10% below nominal, and hours of spells >20% below nominal). Panels A and C of [Figure 5](#) show that the *distribution* of voltage quality across sites was also similar across treatment and control sites prior to the intervention. Monthly nighttime radiance data, a reasonable proxy for power outages, show nearly identical levels and trends for up to 10 years prior to the intervention ([Figure A11](#)).

Taken together, the evidence of baseline balance and parallel pre-trends, in conjunction with the quasi-random institutional design of the line bifurcation investments, supports the logic that a difference-in-differences design will identify the causal impacts of these electricity grid improvements.

²²The p-value for a joint F-test for household characteristics is 0.230, while that for firm characteristics is 0.407.

²³Prior to construction, respondents at control (treatment) sites are on average 233 (253) meters from the nearest transformer (see [Figure A12](#) for the full distribution).

Figure 4: Impacts of transformer injection on voltage by time of day



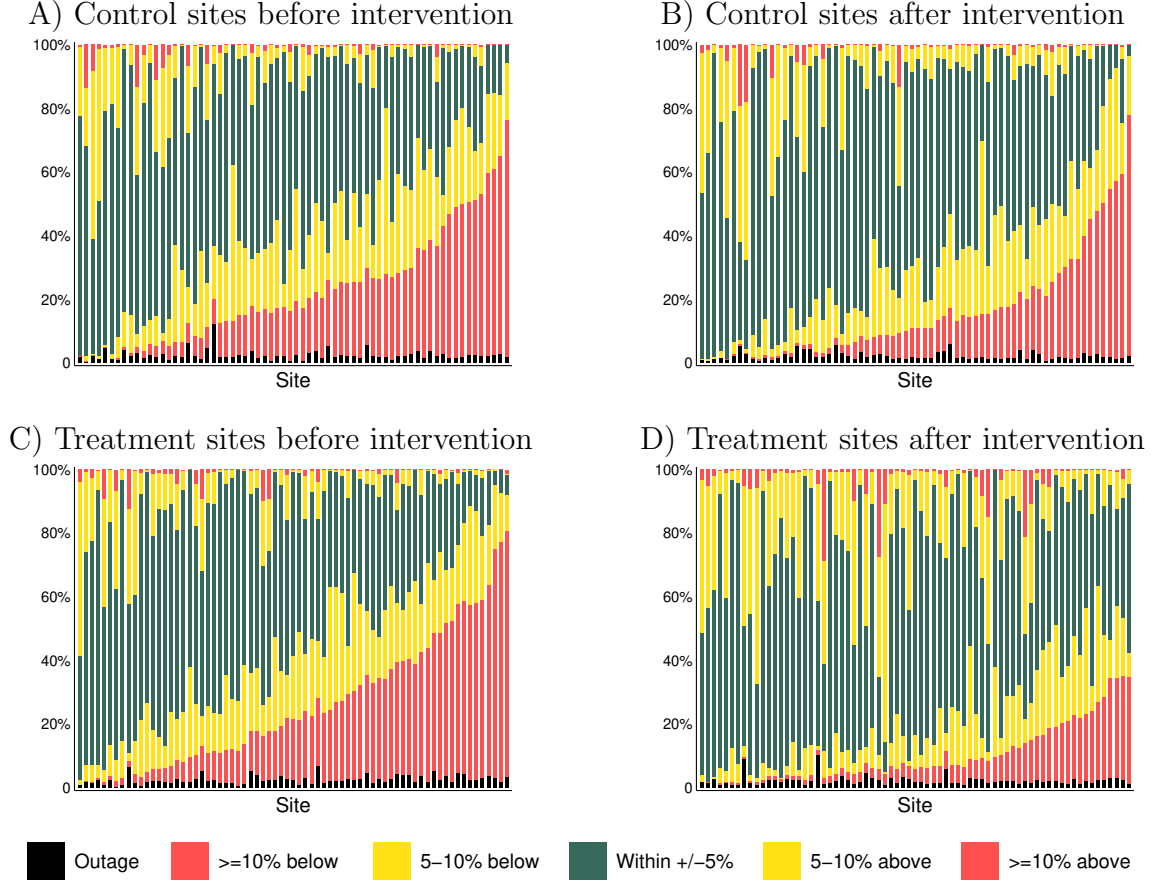
Average voltage by hour of day and treatment status with 95% confidence intervals around treatment means. The dashed line shows Ghana’s nominal voltage (230V). Average voltage increases by 5V in control sites and by 10V in treatment sites after construction. SEs clustered by site. [Figure C2](#) shows impacts on outage duration.

5.4 Results: Impact of line bifurcation on power quality and outages

Injecting new transformers to reduce average loads and distance to transformers should directly increase average voltage. The median respondent in a treatment site was 163 meters closer to the nearest transformer (along the LV network) after line bifurcation – going from 253 to 90 meters away – and there was no change in control sites ([Figure A12](#)). [Figure 4](#) plots average voltage by hour of day and by treatment status before and after construction of new transformers. Panel B shows that the intervention increased average voltage by around 5V in treatment sites relative to control sites across all hours of the day. As a result, average voltage in treatment sites is within $\pm 5\%$ of nominal voltage across all hours.

[Figure 5](#) visualizes the impact of the new transformers by comparing voltage before and after the intervention across control and treatment sites. For each site, it shows the fraction of time the minute-by-minute data are within different voltage ranges. The ordered stacked categories show outages and the fractions of time voltage was at least 10% below nominal, 5–10% below nominal, within $\pm 5\%$ of nominal, 5–10% above nominal, and at least 10% above nominal. The categories are identified by distinct fills in the legend.

Figure 5: Impact of transformer injection intervention on distribution of grid quality



Distribution before and after transformer injection at control and treatment sites. Each stacked bar reports the fractions of time in outage, at least 10% below nominal, 5–10% below nominal, within $\pm 5\%$ of nominal, 5–10% above nominal, and at least 10% above nominal; categories are identified by the legend.

The main effect of the intervention was to reduce the time where voltage was more than 10% below nominal voltage, or between 5–10% below nominal. It also increased the amount of time when voltage was more than 5% *above* nominal voltage, but as discussed in [Subsection 2.1](#), small deviations above nominal voltage are unlikely to negatively affect appliances. The technology cannot capture very brief severe voltage spikes, which can cause serious damage. These may be less frequent for new transformers if the primary cause is mechanical issues with old transformers, but would not have been reduced if they are primarily caused by other power distribution issues. The increase in average voltage thus largely reduces the extent of under-voltage spells, which are more likely than moderate over-

voltage to damage appliances and affect use.

To estimate the causal treatment effect, we use a panel fixed effects regression:

$$Y_{ist} = \beta_0 + \beta_1 \text{Post}_t + \beta_2 \text{During}_t + \beta_3 \text{Treat}_s \times \text{Post}_t + \beta_4 \text{Treat}_s \times \text{During}_t + \mathbf{\Gamma}_s + \mathbf{\Gamma}_t + \epsilon_{ist} \quad (1)$$

Y_{ist} is an outcome experienced by device i in site s at time t , which may be hourly or monthly depending on the analysis. β_1 and β_2 capture changes in overall voltage quality or outages after and during construction relative to the pre-construction period, and β_3 captures the average treatment effect of interest. $\mathbf{\Gamma}_s$ are site fixed effects, which subsume a Treat dummy. $\mathbf{\Gamma}_t$ are time fixed effects that vary across regressions.²⁴ A ‘During’ dummy captures direct effects of construction itself: during this period, temporary influences such as the disconnection of the grid to be able to conduct wiring or the connection of construction machinery could have direct effects. Standard errors are clustered by site in all regressions.

Table 5 presents the results. Columns (1)-(4) use hourly data to estimate the impact of line bifurcation on four metrics of power quality: minutes of power outages, average voltage, the absolute voltage deviation from nominal voltage, and the standard deviation of voltage. In line with Figure 4, Column 2 shows that line bifurcation increased average voltage by 5.5V in treated sites relative to control sites. The pre-construction average voltage was 219V, and the post-construction average rose to approximately 225V in control sites and 230V in treatment sites. We cannot reject equality of the estimated increases in average voltage across devices at different baseline distances from the nearest transformer(Figure C3). The average absolute voltage deviation in a given minute did not change in treatment sites relative to control sites, likely because of increased small deviations above nominal in treatment sites. In addition to improving average voltage, line bifurcation improved the minute-by-minute variation in voltage. Column (4) indicates that after the intervention, the standard deviation of voltage decreased by 0.51 in treatment sites relative to control sites.

²⁴Results are qualitatively unchanged when instead using week-of-sample or day-of-sample fixed effects, which also subsume the Post and During dummies, and when dropping the site fixed effects.

Table 5: Impact of transformer injection intervention on outages and voltage

	(1)	(2)	(3)	(4)	(5)	(6)
	Minutes power out	Average voltage	Absolute voltage deviation	Standard deviation voltage	Hours of spells >10% below nominal	Hours of spells >20% below nominal
During construction	0.21*** (0.07)	0.76 (1.09)	-0.67 (1.19)	0.10* (0.05)	1.13 (10.50)	0.93 (9.37)
Treat X During	-0.06 (0.12)	2.38 (1.60)	-1.47 (1.54)	-0.27** (0.11)	-38.49** (17.57)	-34.37** (15.08)
Post	-0.08 (0.08)	5.94*** (1.74)	-3.74*** (1.14)	-0.18 (0.11)	-43.46*** (14.11)	-32.03*** (11.31)
Treat X Post	-0.21 (0.13)	5.48** (2.48)	-0.83 (1.70)	-0.51*** (0.14)	-54.68*** (20.80)	-41.99** (16.82)
Observations	10033086	9866078	9866078	9831794	14213	14213
Pre-constr. ctl. mean	1.39	219.18	16.51	2.49	125.07	76.69
Hour of day FE	Y	Y	Y	Y	N	N
Week of year FE	Y	Y	Y	Y	N	N
Month of year FE	N	N	N	N	Y	Y
Site FE	Y	Y	Y	Y	Y	Y
Hourly/monthly data	Hourly	Hourly	Hourly	Hourly	Monthly	Monthly

Difference-in-difference results for the impact of treatment on power quality measured by GridWatch devices, with each column a different outcome variable. Columns (1)-(4) use hourly data while Columns (5)-(6) use monthly data. Standard errors are clustered at the site level. [Figure 6](#) and [Figure A10](#) show event study graphs for each outcome variable. [Table C3](#) and [Table C4](#) show alternative specifications considering ways in which transformer injection implementation deviated from initial plans. [Table C5](#) and [Table C6](#) show additional outcomes. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Columns (5) and (6) use monthly data to estimate impacts on low-voltage spells: periods during which voltage drops at least 10% or 20% below the nominal level. The treatment caused a 55 (42) hour decrease in the monthly hours of spells with voltage more than 10% (20%) below nominal, respectively – a 44% (55%) decrease. The number of low-voltage spells per month, defined as periods when the voltage fell at least 10% below nominal, fell by 68 in treatment sites relative to control sites, a 27% decrease relative to the pre-construction control mean. Spells where the voltage drops >20% below nominal decreased by 4.6, or 50% ([Table C6](#)). Remaining spells are not shorter on average in treatment sites, though the maximum monthly spell length is 5.9 hours (50%) shorter after construction relative to control sites. Therefore, although the magnitude of the impact on average voltage may not appear consequential, the size of the impacts on time spent experiencing low voltage

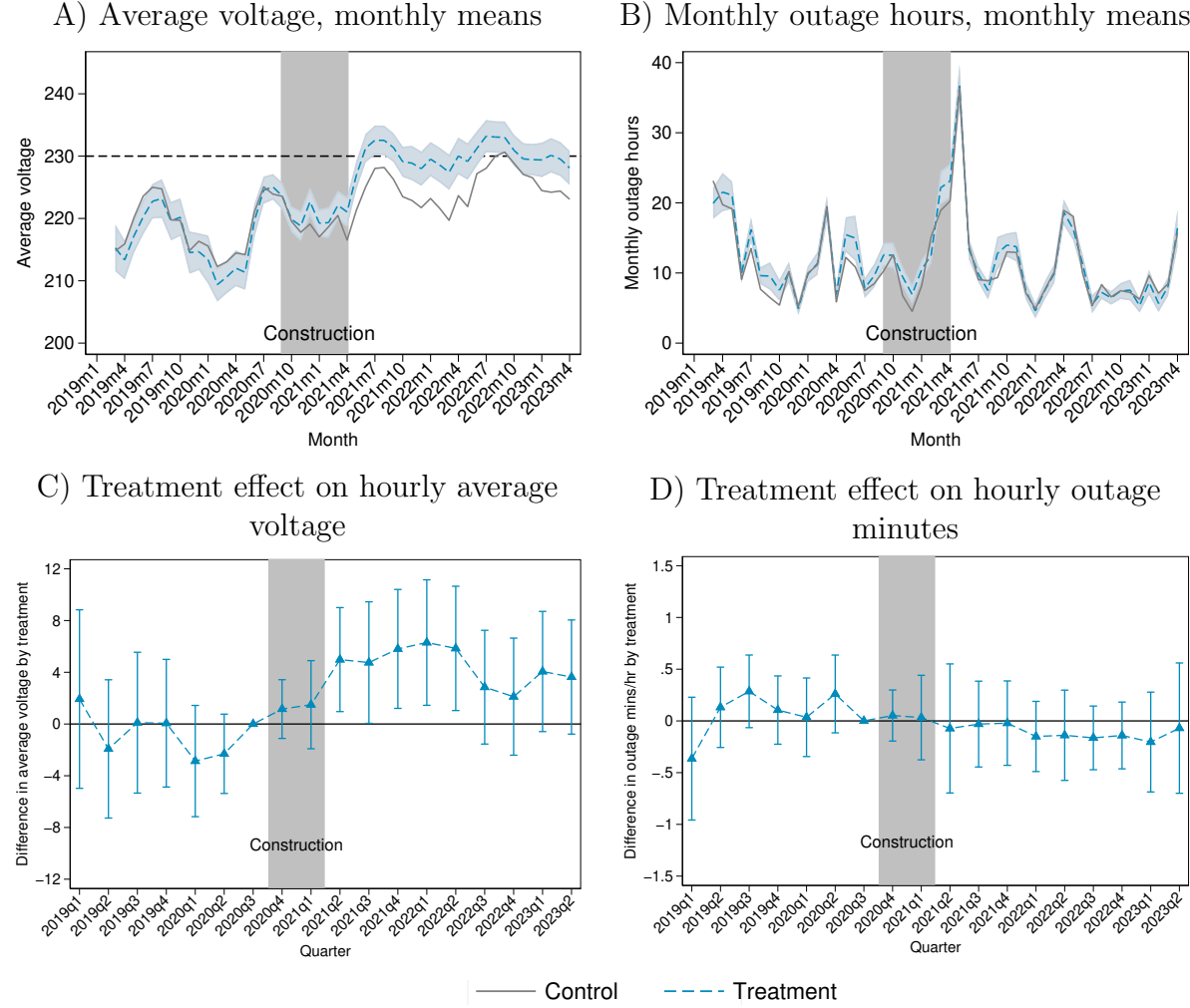
highlights the extent to which the intervention improved power quality.

Reductions in power outages were a central objective of the Ghana Power Compact, with the 19% ex ante estimated economic rate of return relying in part on a 50% reduction in power outages. Yet the effect of the transformer injection intervention on power outages is not statistically significant at conventional levels ($p = 0.11$). One reason for this might be that outages originating in the low-voltage part of the grid – where the line bifurcation work was done – only account for around 30% of all power outages (Panels A and B of [Figure C5](#)) which we infer from the GridWatch device data. The remainder of outages originate in the medium-voltage (such as failures at primary substations or major feeder lines) and high-voltage (such as generation and transmission) parts of the network (Panels C and D of [Figure C5](#)). However, isolating low-voltage outages indicates that there was no significant effect among this subset of outages. Instead, the line bifurcation intervention primarily impacted voltage quality.

[Figure 6](#) presents event study results showing the impact of line bifurcation on average voltage and outage hours by quarter. Outage hours follow similar seasonal patterns before and after construction in both treatment and control sites. Though point estimates of treatment differences in hourly outages minutes post-construction are all negative, none are statistically significant. Panel A shows that average voltage increases post-construction and seasonal fluctuations are mostly smoothed out in treatment sites. Voltage gains at treatment sites relative to control sites are stable for a year after construction completion, but decrease slightly and are no longer significant after around 18 months (Panel C). It is unclear how effects would be expected to evolve over a longer time frame. If consumption rises more in treatment sites in response to improved power quality, the increased load may reduce the voltage gains. On the other hand, as consumption increases across the study area the presence of an additional transformer in treatment sites would reduce the amount of overloading of local networks leading to better long-term voltage quality.

[Figure 4](#), [Table 5](#), [Figure 5](#), and [Figure 6](#) Panel A show that there were voltage quality

Figure 6: Impacts of transformer injection intervention on power quality and reliability



Panels A and B show monthly mean values by treatment with 95% confidence intervals for treatment means. Panels C and D show estimated coefficients and 95% confidence intervals from a regression of outcomes on site treatment status by quarter, controlling for hour of day, week of year, and site fixed effects. SEs clustered by site. [Figure A10](#) shows results for absolute voltage deviation, standard deviation, hours of spells $>10\%$ below nominal, and hours of spells $>20\%$ below nominal. Treatment coefficients use $t = -1$ as the reference period and thus do not exactly match those presented in [Table 5](#), which use the entire pre-period as the reference period. [Figure C6](#) shows Panels C and D without the 25 percent of control sites that saw the most significant improvements in voltage quality. May 2021 saw an unusual level of power outages due to planned construction at two large primary substations (GBC Ghana Online, [2021](#)).

improvements over the study period at control sites (the causal voltage improvement at treatment sites is on top of this improvement at control sites). Broad-scale voltage improvements may be attributable to three factors. First, voltage improvement at control sites could be due to spillover effects from treatment sites, given the physics of power flow across the distribution grid. However, empirical tests indicate that this is likely not the case. The voltage

improvement at control sites does not differ significantly by distance to the nearest treatment site ([Table B4](#)), which one would expect if these improvements were due to reduced load in nearby connected portions of the grid. Furthermore, line bifurcation did not significantly change distances to the nearest transformer in control sites (Panel B of [Figure A12](#)), and combining changes in control site device distances to the five nearest transformers at endline with how those distances correlate with average voltage suggests this would only explain a small share of the control site increase in voltage post-construction.

Second, macroeconomic conditions related to the COVID-19 pandemic might have reduced residential load, contributing to generally improved power quality. However, power quality does not change noticeably with the onset of COVID-19 in 2020, with similar seasonal patterns in average voltage going back to when the first GridWatch devices were deployed in 2018. COVID-19 electricity subsidies in Ghana may have helped to maintain electricity consumption levels despite economic difficulties in this period (Berkouwer et al., [2022](#)). Further, data on electricity spending by survey respondents indicates similar consumption at baseline and endline, indicating changes in voltage quality are not due to changes in electricity loads in the study areas.

Third, and more likely, the broad-scale voltage improvements result from other large-scale concurrent investments that the MCC made in the grid as part of its Ghana Power Compact. The construction of additional primary substations and bulk supply points will have improved the operations of the grid as a whole (MCC, [2014](#)). We return to the discussion of broad-scale voltage improvements, including at control sites, when interpreting the results on customer outcomes below.

In sum, line bifurcation led to improvements in voltage quality along several dimensions without affecting power outages. On average, treatment sites experienced a 5.5 V increase in mean voltage following the intervention. More importantly, line bifurcation reduced the time spent at voltage levels more than 10% and 20% below the nominal voltage (230 V) by 44% and 55%, respectively. With appliances in Ghana largely rated for 220–240 V, this represents

a substantial improvement in voltage quality that likely improves appliance performance and reduces the risk of malfunction or damage associated with sustained low voltage.

5.5 Results: Impact on customer outcomes

We estimate impacts of the line bifurcation treatment on customer outcomes using the following difference-in-differences specification:²⁵

$$Y_{ist} = \beta_0 + \beta_1 \text{Post}_t + \beta_2 \text{Treat}_s + \beta_3 \text{Treat}_s \times \text{Post}_t + \mathbf{X}_i + \epsilon_{ist}, \quad (2)$$

For outcome Y_{ist} experienced by respondent i at site s at time t , which is either the baseline or endline survey period. β_3 captures the differential outcome being observed post-construction in treatment sites – the treatment effect of interest. $\mathbf{X}_i$ are baseline socioeconomic controls.²⁶ Standard errors are clustered by site in all regressions.

Despite a first-stage where voltage quality improves in treatment sites relative to control sites, the treatment has limited effects on outcomes for households and small firms. [Figure 7](#) presents the results for impacts on outcomes related to customer electricity experiences, normalized around the baseline control mean, pooled for firms and households ([Table B5](#) provides detail).²⁷ Reported hours of bad voltage in the last month do not fall significantly in treatment sites, though this can partly be attributed to the timing of the endline survey. Endline data were collected in July-September 2022 at the time of year when voltage is typically better on average, meaning there is less scope for the additional average increase in voltage levels in treatment sites above the increase in control sites to reduce bad voltage

²⁵This specification was registered in our pre-analysis plan (Berkouwer et al., 2019). Voltage improvements were unanticipated so voltage-related analyses were not detailed in the pre-analysis plan. An additional source of identification (spatial discontinuities in feeder connections) did not offer a sufficiently strong first stage and was therefore left to a separate paper. Results for all outcomes listed in the pre-analysis plan are included in [Appendix D](#).

²⁶Age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is a household or a firm, baseline distance to the nearest transformer, and district fixed effects.

²⁷[Table C7](#) shows outcomes for firms alone; the pattern of results is similar.

detected by survey respondents. Indeed, only 8% of respondents report any bad voltage hours in the past month at endline, consistent with this being the study period with the highest average voltage (Figure 6 Panel A), compared to 62% during the baseline in March-April 2021. Seasonality of power quality and the timing of data collection may have affected estimated impacts of the intervention more generally, which we return to below.

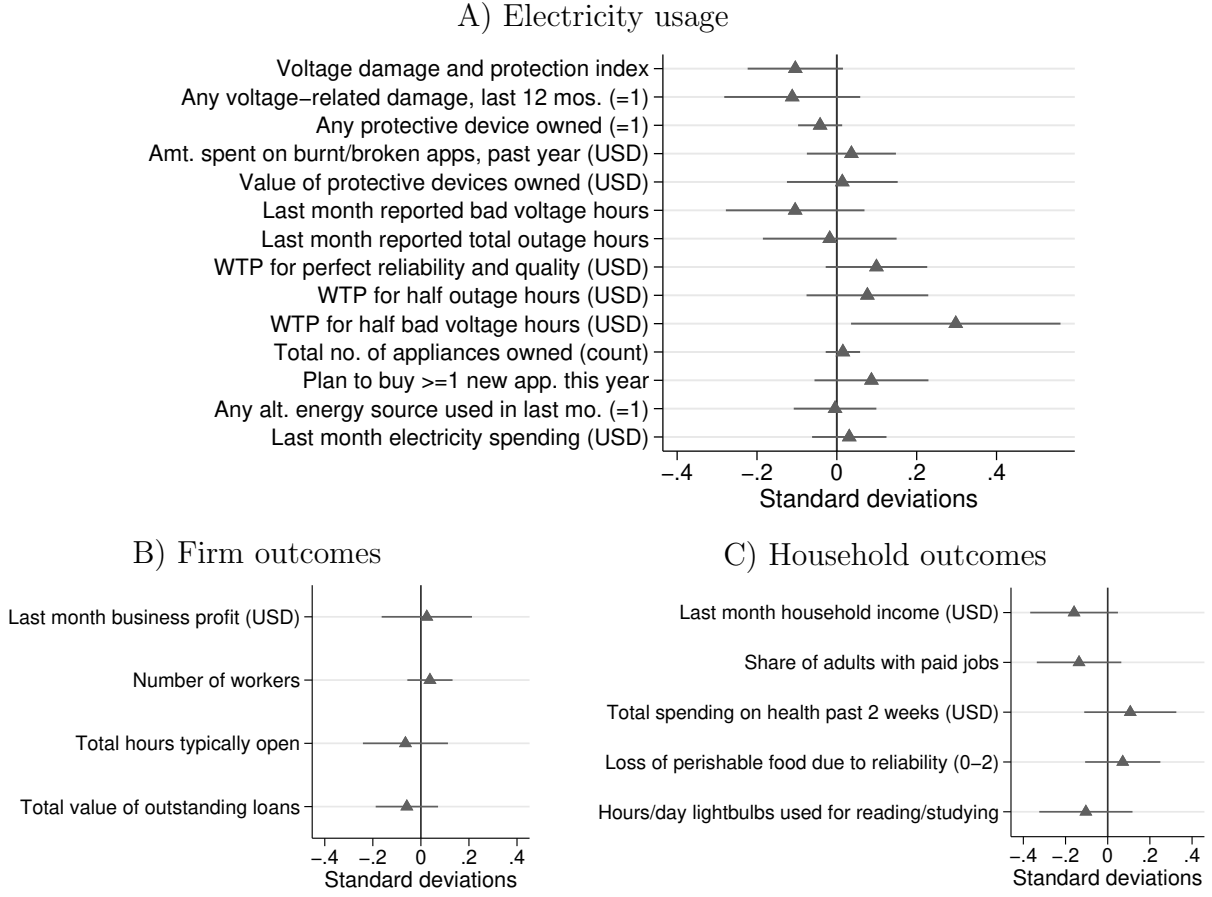
We estimate a 0.1 SD reduction ($p=0.09$) in a voltage damage and protection index comprising two components – whether appliances were damaged by voltage in the past year and whether the respondent has any voltage protective device. The effects on the individual components are noisier and not statistically significant but correspond to a 20% decrease in voltage-related damages and an 8% decrease in the share of respondents reporting having a protective device in treatment sites relative to control sites. This aligns with improved perceptions of voltage quality which may lead households and firms to sell or not replace their protective devices. However, the intervention did not eliminate the risk of voltage-related damages. There is a 20% decrease across the full study area post-construction and a further 20% decrease in treated sites (Table B5), but 20% of respondents in treatment sites still report voltage-related damages occurring after the line bifurcation voltage improvements.

Monthly WTP for an electricity connection with perfect reliability and half the current bad voltage hours increases by \$1 in treatment relative to control sites. WTP for other improved electricity connections does not change significantly. There are no significant impacts on any other electricity-related outcomes. The treatment also has no impact on a variety of firm and household outcomes, as shown in Panels B and C.²⁸

We evaluate potential concerns about our statistical power to detect economic impacts by calculating how large an increase in each outcome can be statistically ruled out relative to the control mean. For many outcomes we could have expected to be affected by the

²⁸Table B6 shows significant decreases in firm costs in treatment sites of a similar magnitude to non-significant decreases in revenues, both of which offset significant increases in control sites, suggesting treatment firms did not grow as much as control firms. These effects do not survive adjustments for multiple hypothesis testing (Table C8). Table B7 shows no significant effects on a number of additional household outcomes.

Figure 7: Impact of transformer injection treatment on primary outcomes



Coefficients and 95% confidence intervals for β_3 ($\text{Treat}_s \times \text{Post}_t$) from Equation 2. All units are standard deviations relative to the baseline control mean. Panel A pools firm and household respondents. The voltage damage and protection index comprises ‘any voltage-related damage’ and ‘any protective device owned’. Table B5 shows the results for non-normalized versions of these and additional outcomes related to electricity. Table B6, and Table B7 show results for non-normalized versions of the outcomes in for firms in Panel B and households in Panel C, respectively, along with additional firm and household outcomes.

voltage improvements, including business revenue, household income, and employment, we can rule out relative increases of less than 10% with 95% confidence. Other outcomes with more baseline variance such as business profits and loss of perishable foods have less precise treatment estimates and we cannot rule out larger increases. In general the high level of precision for outcomes with tighter baseline distributions suggests that our null results are not primarily driven by a lack of statistical power. Instead, they indicate the absence of substantial economic impacts across the outcomes studied. We discuss potential reasons for these null effects below.

The improved power quality in control sites stemming from broad-scale electricity grid investments is reflected in significant estimated average differences between baseline and endline electricity outcomes ([Table B5](#)). By the endline period, all sites saw declines in the probability of having had an appliance damaged by voltage issues over the past 12 months, spending on damaged appliances, ownership of voltage protective devices, reported hours of outages and bad voltage, and WTP for improved electricity connections. Self-reported daily hours of bad voltage were nearly zero during the endline even at control sites. Seventy-one percent of respondents at endline said voltage was much better than two years ago (19% said it was slightly better), and firms were 19 percentage points less likely to report bad voltage as an obstacle to firm operations at endline compared to baseline. Improvements in reported electricity outcomes across the whole sample may also have been due to seasonal differences in energy use and economic activity due to the differences in the timing of the two surveys, with the endline survey coinciding with the time of year when voltage is typically better.

Our findings of null treatment effects on most customer outcomes are robust to a variety of potential sources of bias ([Appendix C](#)). There is no evidence of treatment spillovers ([Table B4](#)). Still, to address potential SUTVA violations due to any potential treatment spillovers, we drop control sites closer to treatment sites than median (1.3km). We also drop ‘movers’ and anyone for whom the monotonicity assumption on distance to the nearest transformer is violated ([Table C9](#)). These changes to the sample do not change the results. We next consider differences across treatment sites in where and when planned transformer construction was completed, drawing on contractor reports and our own monitoring visits ([Table C3](#), [Table C4](#), and [Table C9](#)).²⁹ Increases in average voltage and decreases in voltage damages are slightly larger in sites where in-person visits confirmed the presence of new transformers in treatment sites. However, we cannot reject equality of coefficients, and even in sites with confirmed transformer injections we find null effects on most socioeconomic

²⁹We drop two treatment sites where the contractor indicated that the transformer was not commissioned and drop additional treatment sites where additional construction monitoring found no new transformer constructed. We also run an instrumental variables version of this regression, using treatment assignment as an instrument for new transformer construction.

outcomes.

5.6 Illustrative benefit-cost implications

Although we do not observe short-run impacts of the line bifurcation investments, the welfare implications of reduced appliance damage and defensive investments can still be approximated. To illustrate the potential magnitude of these benefits, we present a simple back-of-the-envelope calculation based on the observed reductions in voltage-related appliance damage and protective investments.

MCC spent \$13.9 million on low voltage line bifurcation in the Achimota, Dansoman, and Kaneshie districts of Accra, which had an estimated combined population of around 49,000 households (GSS, 2014), or \$286 per household. While we do not find statistically significant short-run impacts on economic outcomes, we can bound a meaningful share of the potential returns by calculating an implied benefit-cost ratio based on baseline expenditures on protective devices and appliance repairs.

The average customer holds \$14 worth of voltage protective devices and spends \$11 annually on repairs and replacements of broken appliances. Even if the intervention eliminated all protective devices and 100% of damage repair/replacement expenditures every year for 30 years (discounting at $\delta = 0.9$) for all 49,000 households, the total economic benefit would be \$11.6 million, still falling short of the investment cost: the benefit-cost ratio is 0.83.³⁰ Based on the outcomes we directly observe in the timeframe of this study (smaller and noisier effects on appliance damages and protective device ownership), the line bifurcation investment does not appear to generate benefits in excess of its costs.

While the intervention likely will not eliminate *all* voltage-related damages, these impact estimates could nevertheless be a significant under-estimate of total economic benefits if additional socioeconomic impacts emerge in the longer term. Customers will also benefit

³⁰The estimated benefit-cost ratio is sensitive to the assumed discount rate. Using an 8% annual discount rate instead of 10% increases the benefit-cost ratio to approximately 1.0, while a 5% discount rate increases it to 1.37.

from increased utility from their appliances, and from a reduced risk of harm from burning or fires from appliances affected by severe voltage fluctuations. These back-of-the-envelope calculations furthermore do not include benefits to firm customers and to the electricity utility through reduced technical losses and maintenance costs.

An alternative policy directly responding to voltage quality issues could be to directly subsidize customer purchases of voltage protective devices. We estimate that providing one protective device (median value \$24) to every household in the study districts would cost approximately \$1.18 million, less than one-tenth the cost of the infrastructure upgrades. The true cost of directly protecting all meaningful appliances in the study area would likely be much higher, as most customers own several valuable appliances (such as a television and fridge for households), and businesses would also require protective devices. While such a policy could more cheaply reduce costs from appliance damages, repairs, and health risks than line bifurcation, it would merely treat the symptoms of poor voltage quality. Stabilizers and similar devices protect specific appliances in response to a failing system but do not improve the utility of the electricity itself by increasing the amount of time during which appliances can be productively used. Therefore, while a protective device subsidy program would be less costly than grid investments to improve voltage, its potential benefits would also be significantly smaller.

Calculating expected long-term benefits of line bifurcation based on avoiding appliance damage costs at baseline may also underestimate the full amount of avoided damages. Electricity consumption is projected to increase in response to increasing incomes and temperatures, raising loads on distribution transformers. In the absence of new transformer injections, increased loads could worsen voltage quality issues, causing more damage to appliances and human users and more costs from reduced appliance usability (Berkouwer et al., [2026a](#)). By improving distribution capacity through line bifurcation, such risks may be mitigated. Moreover, if appliance ownership and electricity use expand gradually as reliability improves, the resulting reductions in damage and disruptions could generate larger and

more significant economic benefits over longer horizons. In this sense, the line bifurcation investments provide "option value" for future growth that would not be generated by policies targeting the symptoms of bad voltage alone.

Ultimately, the justification for the \$13.9 million investment likely rests not solely on savings from appliance damages and protection, but on long-term grid resilience, reduced health risks, and benefits to appliance productivity. These dimensions are difficult to value but are critical to the broader economic return on infrastructure.

6 Discussion: Explaining the null economic impacts

These findings raise the question: why do voltage improvements not translate into more substantial economic impacts on customers in the 16 months after the intervention? Voltage quality improved significantly in treatment sites relative to control sites. Voltage instability is a clear concern for consumers, leading to appliance damage and a willingness to pay for better power quality, as shown in [Subsection 4.1](#). Despite this, the treatment has minimal socioeconomic effects between baseline and endline. We consider several possible explanations.

6.1 Is treatment intensity too small to detect socioeconomic impacts?

One possible explanation for no significant socioeconomic impacts is that the net improvement in voltage quality in our main specification – approximately 5.5V on average – may be too small to generate detectable economic effects. If the treatment intensity is limited, even meaningful relationships between voltage quality and economic outcomes may not be observable in the baseline treatment-control comparison. Therefore we examine whether larger improvements in voltage are associated with stronger economic effects.

First, we assess treatment effect heterogeneity by baseline voltage quality. We categorize

sites according to whether they are above or below the median daily hours that voltage was within 10% of nominal at baseline. Sites with below median baseline voltage (averaging 210V at baseline) experienced larger improvements, reaching an average of 220V at control sites and 229V at treatment sites at endline (Table B8). In contrast, sites with above median voltage (averaging 228V at baseline) experienced only small and statistically insignificant increases in average voltage. The difference in the effect on power quality is illustrated more sharply when considering the share of time that voltage is at least 10% below nominal, or below 207V (Table B9). Sites with worse baseline voltage were at least 10% below nominal 36% of the time. While this fell to 23% of the time in control sites at endline, it fell to just 4% of the time in treatment sites. Sites with better power quality at baseline were 10% below nominal 7% of the time on average and experienced no significant changes at endline.

These differences in voltage improvements between the two categories of sites are economically meaningful in the Ghanaian context. As discussed earlier, many appliances in Ghana are rated for 220–240V, making them more vulnerable to moderate voltage fluctuations than appliances used in higher-income settings that tolerate a wider range of input voltages. As a result, the observed first-stage voltage improvements due to the line bifurcation treatment represent a more substantial change in effective power quality for the below-median sites than for the above-median sites. If voltage improvements have non-linear impacts, larger treatment effects should therefore appear among sites with lower baseline voltage.

However, we find little difference in treatment effects across the two groups. Even among sites with below-median voltage at baseline, the estimated socioeconomic impacts remain largely statistically insignificant (Table B10, Column 1).

A second approach is to remove control sites that experienced large increases in voltage between baseline and endline from the sample. We restrict the sample of control sites to those that experienced less than 10V of improvement in average voltage after line bifurcation, which is the 75th percentile of voltage improvement in the control sites. The remaining control sites and treatment sites are balanced in terms of observable characteristics at baseline

(Table C11). This increases the difference between treatment and control sites to more than 10V (Figure C6, Table C12). Still, even among this subsample, the estimated effects on firm and household outcomes are largely unchanged (Table C13, Table C14). Together, these results indicate that a larger increase in voltage in treatment sites may not have caused larger socioeconomic impacts.

6.2 Persistent outages overshadowing voltage improvements?

Another possibility is that persistent outages might overshadow voltage improvements. As we show in Subsection 5.4, line bifurcation improved voltage quality without affecting power outages. While less severe than Dumsor, outages were still common during our sample period. Table 1 shows that the average respondent experienced 14 hours of outages per month. With this level of consistent exposure to outages, even large voltage improvements may be perceived as insufficient to induce meaningful changes in economic behavior.

To address this concern, we test for heterogeneity in treatment effects by baseline (and endline) outage levels using both device- and respondent-reported outage data. Specifically, we estimate differences in average treatment effects between respondents with below- and above-median outage hours. If persistent outages overshadow improvements in voltage quality, we would expect larger effects among respondents with better average power reliability. However, the results in Table B11 do not support this hypothesis. We find no statistically significant differences in the effect of line bifurcation on economic outcomes between individuals with above- and below-median pre-treatment monthly outage hours. The same is true when testing differences by post-treatment outage hours and when considering reported outages as opposed to those detected by GridWatch devices.

6.3 Heterogeneous treatment effects by the energy intensity of the customer?

Power quality is especially important for households and businesses that depend heavily on electricity for production and daily activities, implying that improvements in voltage may have larger effects for these customers. Our study focuses on households and micro-enterprises, which are typically less energy-intensive than larger firms. As a result, improvements in voltage quality could plausibly generate larger economic impacts for more electricity-intensive firms than those represented in our sample. In related work, Chaurey and Le (2022) show that rural infrastructure improvement programs in India, which included investments in electricity distribution that reduced outage-related problems, increased employment particularly in electricity-intensive industries.

In our setting, the areas targeted for the line bifurcation investments did not include many wealthy households or medium to large businesses, who may have experienced larger treatment effects. While we cannot directly test whether economic impacts would have been larger for such customers, we test for differences within our sample by baseline characteristics that reflect the importance of voltage quality for respondents, including reported electricity importance (for firms), willingness to pay for perfect electricity reliability and quality, ownership of protective devices, general appliance ownership, and fridge ownership. In general we find no significant differences (Table B10 Columns 3–7).

Though the focus on micro-enterprises in the sample may limit the generalizability of our findings, particularly to larger firms which might have benefited more from voltage improvements, these small firms account for a large share of employment and income generation in emerging economies. Furthermore, evidence from other contexts shows that improvements of power quality can affect households and small firms (e.g., Chaurey and Le, 2022; Meeks et al., 2023).

We also identify a set of respondents who may be more susceptible to voltage issues: those who experienced below median voltage at baseline and at the same time owned appliances

sensitive to voltage but no protective devices. There are similarly no significant differences in this 14% of the sample that had the “worst” baseline conditions ([Table B10](#) Column 8). These findings suggest that the absence of short-run socioeconomic impacts is not driven by heterogeneity in electricity dependence within the population of households and micro-enterprise population we study.

6.4 Inertia in appliance adoption and potential for longer-term impacts

One explanation for the absence of short-run socioeconomic effects is delayed adjustment along the durable capital margin. Voltage improvements reduce appliance damage and defensive expenditures immediately as shown in [Subsection 5.5](#), but firms and households may postpone new capital purchases until reliability improvements appear sustained.

More generally, the absence of short-run socioeconomic effects following the line bifurcation intervention could reflect two possibilities. Improvements in voltage quality may have limited economic value, or households and firms may delay durable capital investments until improvements in electricity reliability are perceived as sustained. We show above that voltage issues create meaningful economic costs and that customers are willing to pay to reduce those costs, suggesting improvements would have economic value. But in environments with a long history of unreliable service – characterized by both outages and sustained under-voltage – customers may rationally maintain a limited stock of electric appliances and postpone expansion until reliability improves in a durable way. If economic gains from voltage improvements depend on capital investments in appliances and machinery, inertia in investment could explain the null economic effects we observe.

Ghana has faced over a decade of unreliable electricity supply, marked by both outages and low voltage, with intermittent periods of improvement. Customers may not interpret observed improvements in voltage quality as sufficient evidence that electricity supply has become reliably usable for new capital investments. We find evidence consistent with inertia

in perceptions of overall electricity conditions. Respondents do report reductions in bad voltage hours following the line bifurcation treatment, indicating they recognize a power quality improvement. But we observe no impact of the treatment on customers' willingness to pay for improved electricity connections (Figure 7), and treatment does not affect perceptions about the likelihood that overall power quality and reliability have improved or will improve in the next year (Table B7). These null effects may reflect customer uncertainty in the persistence of the voltage improvements they observe.

In such a context, households and micro-enterprises may be delaying new capital investments in appliances until improvements in power quality are perceived as more permanent. The relatively short follow-up period – approximately 16 months after the intervention – may therefore be insufficient for customers to adjust their appliance stocks. Broader economic disruptions during the study period, including those associated with the COVID-19 pandemic, may also have contributed to temporarily suppressed household investment during this period.

Consistent with this interpretation, treatment did not increase household appliance adoption in the near term. At baseline the median respondent owned three different types of electric appliances along with light bulbs. We observe no difference in appliance acquisition after line bifurcation by treatment status: 37 percent of respondents at endline in both treatment and control sites reported acquiring at least one new appliance since the baseline survey, with no difference between firms and households (Figure A13). Many of these new appliances replaced older units rather than expanding the total stock, such that the mean total count of electric appliances did not change significantly between baseline and endline. This pattern suggests that improvements in voltage quality *reduced damage and facilitated replacement*, but did not yet induce expansion of appliance capital.

However, there is suggestive evidence that customers in treatment sites may be more likely to expand appliance ownership over time. Fifty-six percent of the sample at both baseline and endline report plans to purchase at least one new appliance in the next year.

Customers in treatment sites are 4.3 percentage points (7.4 percent) more likely to state that they plan to purchase a new appliance following the line bifurcation intervention, although this estimate is statistically imprecise (Figure 7; $p = 0.232$). While these patterns are consistent with delayed adjustment along the durable capital margin, our data do not allow us to directly observe the timing of appliance adoption beyond the 16-month horizon.

Taken together, the evidence in this section is consistent with inertia in customers' investment decisions as an important explanation for the lack of short-run socioeconomic impacts of line bifurcation. If appliance investment responds only gradually to improvements in reliability, short-run estimates may understate the long-run economic returns to distribution upgrades. However, we emphasize that this interpretation is suggestive rather than conclusive, as several of the coefficient estimates underlying it are imprecisely estimated (e.g., the effect on plans to purchase a new appliance in Figure 7). At the same time, this interpretation is in line with previous evidence that shows appliance adoption in Africa is slow, especially in settings with a long history of unreliable service (Dinkelman et al., 2024).

7 Conclusion

Global energy policy in low- and middle-income countries has thus far placed limited focus on the role of voltage quality for economic activity, in part due to data limitations for measuring both the severity of voltage problems and the impact on households and firms. We analyze 337 million temporally and spatially high-frequency power quality measurements, as well as panel surveys with 2,000 households and firms, to generate some of the first evidence on the large-scale economic impacts of voltage quality problems.

Poor voltage quality is a pervasive problem: average voltage is 219V – well below nominal voltage of 230V – and is more than 10% below nominal over 20% of the time. These issues create real economic costs for customers in terms of appliance damages and interference with firm operations. Households and firms are willing to pay similar amounts for electricity

connections with reduced outages and with improved voltage, indicating that poor electricity quality imposes costs on customers that are of a similar order of magnitude as the costs from poor electricity reliability.

An intervention that added transformers to the grid increased average voltage by 5.5V and modestly reduced voltage-related damages and ownership of protective devices. However, the investment had no detectable impact on household and firm outcomes such as electricity spending, appliance ownership, firm profits, and household income. There are no significant differences in treatment effects across different subgroups, even those that ex-ante are more likely to be impacted by an improvement in voltage quality though the sample does not include larger businesses and wealthy households, who may have greater dependence on electricity.

We show that the null economic effects are not due to a weak first stage: even a larger voltage quality improvement has little economic impact. We also show that the null effects cannot be explained by persistent outages or spillovers to control sites. Instead, we show that the lack of economic effects despite results highlighting the self-reported importance of power quality to customers may partly be due to customers' investment inertia. We find suggestive evidence of improvements in mechanical outcomes: customers report reductions in voltage related issues and ownership of voltage protective devices. This indicates that voltage improvements were salient to customers. However, within 16 months of line bifurcation, we observe no effect on appliance acquisition, which is a direct channel through which voltage improvements can impact economic activity. Consistent with investment inertia, we find that respondents in treatment-sites are 7.4 percent more likely to report plans to purchase a new appliance in the subsequent year, although the estimate is statistically imprecise. These results are in line with previous research that highlights the slow appliance adoption in settings with a long history of unreliable service. Treatment effects may be realized in the longer term as perceptions of improved voltage translate into greater appliance ownership and use. The distribution of load across additional transformers also provides a foundation of

stronger power quality as rising incomes and temperatures increase electricity consumption and loads on local electricity networks.

These results present novel evidence on the economic costs of voltage quality, while at the same time highlighting that realizing economic gains through voltage improvement may require longer time horizons and complementary shifts in appliance ownership. We offer a framework with which to evaluate voltage quality investments. Resource-constrained governments should evaluate these electricity distribution infrastructure costs within a broader strategic framework that accounts for long-term growth in energy consumption, grid resilience, and the cumulative costs of power quality degradation over the life of the assets.

Data availability: A replication package with all data and code needed to replicate the analysis in this paper is available on the [JPE:Micro Dataverse](#) (Berkouwer et al., 2026b).

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